

Can Machine Learning Outperform Professional Consensus Forecasts in Forecasting Emerging-Market Sovereign Yields?

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Abstract

This paper examines whether machine learning can systematically outperform professional consensus forecasts in emerging-market sovereign yield prediction. Using a panel of 3,662 observations across 16 emerging markets (2009 Q1–2025 Q2), drawn from an ongoing research project by Chi Chun-Che and Kenji Wada, we compare ARX, LightGBM, and LSTM against the consensus benchmark within a forecast error correction framework.

All three models statistically significantly outperform the consensus (Clark-West test, $p < 0.01$), with overall OOS R^2 of 0.374, 0.490, and 0.470 respectively. The central finding is that different maturities are governed by different learnable structures: 3M yield errors are dominated by linear overreaction (ARX strongest); 2Y errors by cross-period temporal dynamics (LSTM strongest, OOS $R^2 = 0.794$); and 10Y errors by static non-linear interactions (LightGBM strongest). Machine learning gains are largest in moderate stress regimes (VIX between 17 and 23) and deteriorate under extreme crisis conditions. Based on these findings, this paper proposes a VIX-triggered model selection rule for practitioners.

Keywords: emerging-market sovereign yields, machine learning, consensus forecasts, forecast error correction, LightGBM, LSTM, out-of-sample forecasting, Clark-West test, SHAP analysis

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1 Introduction

Forecasting sovereign bond yields in emerging markets is a fundamentally multi-layered problem. Yields are jointly determined by domestic macroeconomic conditions, the global interest rate environment anchored by U.S. monetary policy, and time-varying global risk sentiment that can abruptly reprice sovereign risk irrespective of country fundamentals. This complexity has sustained a large empirical literature on the determinants of emerging-market sovereign yields and spreads, but has also made accurate forecasting elusive—even for professional forecasters who continuously synthesize publicly available information.

The central question of this paper is not whether machine learning can forecast sovereign yields in absolute terms, but whether it can systematically add predictive value *beyond* the professional consensus forecast. This framing is economically meaningful for two reasons. First, professional consensus forecasts represent a strong practical benchmark: they aggregate the views of a large number of institutional analysts who have continuous access to market data, proprietary models, and macroeconomic expertise. If a machine learning model cannot improve upon this benchmark out of sample, its practical value is limited. If it can, the result would imply that consensus forecast errors contain systematic, learnable patterns—patterns that reflect either the bounded rationality of individual forecasters or the inability of standard forecasting frameworks to capture nonlinear interactions among macro-financial variables.

Second, framing the problem as forecast error correction rather than yield-level prediction has methodological advantages. It cleanly separates the question of whether machine learning adds *incremental* value from the harder question of whether it can replicate the full information set of professional forecasters. The natural benchmark—predicting the forecast error as zero, i.e., trusting the consensus unconditionally—is both theoretically principled and practically relevant.

This paper asks the following research question:

Can machine learning outperform professional consensus forecasts in forecasting emerging-market sovereign yields across maturities and forecast horizons?

We study this question using a panel dataset covering **16 emerging-market economies** across Central and Eastern Europe, Southeast Asia, Latin America, and other Asia, spanning **2009 Q1 to 2025 Q2**. The panel is organized at the country $\times$ tenor $\times$ forecast date $\times$ horizon level, yielding 3,662 usable forecasting observations across three maturity buckets (3M, 2Y, 10Y) and six forecast horizons (3, 6, 9, 12, 18, and 24 months). We compare three estimated models—an autoregressive model with exogenous variables (ARX), a gradient-boosted tree model (LightGBM), and a long short-term memory network (LSTM)—against the professional consensus benchmark.

The paper makes two primary contributions. First, it applies machine learning to a practically important forecasting problem in emerging-market sovereign debt, using a benchmark that is both strong and economically meaningful. Unlike prior work that forecasts yield levels from macroeconomic predictors alone, we directly test whether machine learning can identify systematic biases in professional forecasts—a harder and more practically relevant test. Second, we examine *when* machine learning is most useful: we evaluate forecast performance across maturities, forecast horizons, and crisis versus non-crisis regimes, and conduct an ablation study on training window length to

assess the sensitivity of machine learning performance to the information horizon available at each forecast date.

Two methodological contributions accompany the empirical analysis. We explicitly document our strategy for handling the severely unbalanced panel—in which the 10Y segment accounts for approximately 80% of usable observations while the 3M segment contributes less than 8%—and justify a pooled estimation approach with maturity and horizon fixed effects as the primary design. We also provide a systematic comparison of rolling window configurations, showing how the choice of training window length affects out-of-sample performance across horizons and models.

The forecasting panel used in this paper is drawn from an ongoing research project by Chi Chun-Che and Kenji Wada. The dataset is a long panel of emerging-market sovereign yield observations linked to professional consensus forecasts, realized yield outcomes, domestic macro forecasts, U.S. Treasury yield consensus forecasts, global financial market variables, and crisis indicators, all sourced from globally recognized financial databases. We gratefully acknowledge Chi Chun-Che and Kenji Wada as the original authors of this dataset.

The remainder of this paper is organized as follows. Section 2 reviews the related literature. Section 3 describes the data. Section 4 presents the methodology, including the forecasting framework, panel structure treatment, rolling window design, and model specifications. Section 5 reports the main empirical results. Section 6 discusses the findings, and Section 7 concludes.

2 Related Literature

This paper connects four strands of literature: the determinants of emerging-market sovereign yields, machine learning applications to bond yield forecasting, systematic biases in professional forecasts, and out-of-sample forecast evaluation methodology.

Determinants of emerging-market sovereign yields: A large body of work establishes that emerging-market sovereign bond yields are jointly driven by domestic fundamentals and global financial conditions. Csonó and Ivashenko (2013) analyze the dynamics of emerging-market sovereign spreads and find that, while country-specific factors such as fiscal balances, GDP growth, and debt dynamics play an important role in the long run, global factors—particularly U.S. interest rates and the VIX—dominate sovereign risk pricing in the short run, with economies exhibiting stronger fundamentals displaying lower sensitivity to shifts in global risk sentiment. Jaramillo and Weber (2013) further show that the transmission of global shocks to domestic bond yields in emerging markets is state-dependent: when global risk appetite is low, domestic yields are predominantly driven by inflation and real GDP growth expectations, whereas in tranquil times market pricing places greater weight on idiosyncratic fundamentals. More recently, IMF (2025) documents that a ten-percentage-point increase in the VIX is associated with economically significant yield spread widening across a broad panel of emerging markets, with the effect amplified in countries with weaker fiscal and external buffers. These findings motivate the inclusion of U.S. rate factors, VIX, and exchange rates in our forecasting framework and suggest that systematic forecast errors are likely to arise precisely when global financial conditions shift in ways that are difficult to anticipate using standard forecasting models.

Machine learning for bond yield forecasting: The use of machine learning in fixed income forecasting has grown substantially, but most existing work focuses on forecasting yield *levels* rather than correcting for systematic biases in professional forecasts. Zhou (2024) applies random forest and gradient boosting to forecast 10-year local-currency sovereign yields in 17 emerging-market economies, finding that tree-based models outperform linear benchmarks in capturing nonlinear relationships among macro-financial predictors.¹ Nunes et al. (2026) propose an LSTM-LagLasso approach to bond yield forecasting that captures long-term temporal dependencies and improves interpretability relative to standard neural networks. Li, Wang, and Wu (2025) compare five models—linear regression, Lasso, random forest, XGBoost, and SVR—for forecasting the U.S. 10-year Treasury yield using lagged yields, the dollar index, S&P 500, GDP, and PCE as features, finding that XGBoost achieves the best performance in terms of MAE and accuracy, followed by random forest. In the Taiwanese market, Yuan (2018) applies random forest to forecast the Taiwan 10-year government bond yield, confirming that the U.S. 10-year Treasury yield is among the most important features; however, the evaluation relies on random train-test splitting rather than rolling out-of-sample prediction, which may overstate model performance in a financial time series context.

Two limitations are common to all of the above studies, both of which our paper addresses directly. First, these studies benchmark against simple statistical models—such as linear regression or a random walk—rather than against the professional consensus forecast, which is a substantially stronger

¹Zhou (2024) is published in *Highlights in Business, Economics and Management*, a conference proceedings volume. We include it as the closest existing study on ML forecasting of emerging-market sovereign yield levels, while noting that the evaluation design differs materially from ours.

comparison standard because it already embeds the information and judgment of a large number of institutional analysts. Second, the target variable is yield *levels* rather than *forecast errors*, which means these studies do not directly address whether machine learning can add incremental value beyond what professional forecasters already provide.

Systematic biases in professional forecasts: A distinct strand of literature examines whether quantitative models can identify and correct systematic errors in expert forecasts. Bianchi, Ludvigson, and Ma (2022) combine a data-rich environment with a machine learning algorithm to estimate time-varying systematic expectational errors—“belief distortions”—embedded in survey responses. They find sizable distortions even for professional forecasters, with all respondent types overweighting the implicit judgmental component of their forecasts relative to what can be learned from publicly available information; forecasts of inflation and GDP growth oscillate between optimism and pessimism by large margins, with distortions evolving dynamically in response to cyclical shocks. This directly supports the central premise of our paper: that consensus forecast errors contain systematic, learnable patterns. Bianchi, Lee, Ludvigson, and Ma (2025) extend this framework to equity markets, introducing the concept of “prestakes”—predictable mistakes that are especially prevalent during periods of market turbulence—and showing that machine learning algorithms, by independently attending to fundamentals, credit risk, and sentiment, make abrupt course corrections at critical junctures while human forecasters tend to anchor on recent local trends. We apply the same logic to emerging-market sovereign yields: if consensus forecasters make systematic errors when global financial conditions shift, machine learning should exhibit its largest incremental predictive value precisely in high-stress regimes—a hypothesis that motivates our crisis versus non-crisis subgroup analysis.

Out-of-sample forecast evaluation: Our evaluation design follows established principles for time series forecasting. We adopt rolling out-of-sample prediction rather than random train-test splitting, ensuring that each forecast uses only information available up to the forecast date and thereby mimicking the information environment of an actual forecaster. We report the out-of-sample R^2 (OOS R^2) as the primary accuracy metric, defined as the proportional reduction in mean squared error relative to the professional consensus benchmark; a positive value indicates improvement over the consensus, and a negative value indicates deterioration. In the broader macroeconomic forecasting literature, Goulet Coulombe et al. (2022) show that when the predictor space is large and interactions among variables are complex, density-based machine learning methods such as gradient boosting outperform sparsity-based models such as Lasso, especially at short horizons—a finding that motivates our choice of LightGBM as the primary nonlinear model. In the context of forecasting competitions, Makridakis, Spiliotis, and Assimakopoulos (2022) report that in the M5 Accuracy competition, all top-50 submissions were pure machine learning methods, with LightGBM used by all top-50 competitors and outperforming all statistical benchmarks by a substantial margin.

3 Data

3.1 Sample Construction

The forecasting panel used in this paper is drawn from an ongoing research project by Chi Chun-Che and Kenji Wada. The dataset is a long panel of emerging-market sovereign yield observations linked to professional consensus forecasts, realized yield outcomes, domestic macro forecasts, U.S. yield consensus forecasts, global financial variables, and crisis indicators, all sourced from globally recognized financial databases. We gratefully acknowledge Chi Chun-Che and Kenji Wada as the original authors of this dataset.

The construction of this forecasting panel involved multiple layers of data engineering work spanning approximately nine months. The raw candidate sample contains 154,825 observations, of which 3,662 satisfy the joint requirement of a non-missing consensus forecast and a non-missing realized yield outcome. The core challenge in panel construction was identifying the actual bond instrument underlying each consensus forecast in each country-quarter combination. The benchmark yield index maintained by the globally recognized financial database does not track a fixed bond instrument; rather, it periodically rolls to a different actual bond as market liquidity conditions evolve—for example, consensus forecasts for Bulgaria’s 10-year sovereign yield may reference different actual bond instruments across quarters, even though the forecasting target remains consistently the 10-year sovereign bond of that country. This paper resolves this issue by identifying, quarter by quarter, the actual bond instrument referenced by the generic yield index for each country-tenor pair, and using the corresponding realized yield-to-maturity to compute forecast errors. In addition, each quarterly snapshot in the consensus forecast database is a static record representing forecasts submitted by institutional contributors prior to the survey cutoff date; this paper treats each observation as a forecast formed at a specific forecast date for a specific target date, rather than treating snapshots from different cutoff dates as a continuous time series. The data integration and cleaning pipeline was implemented in Excel VBA for initial field extraction and ticker identification, and subsequently in R for panel merging, sample selection, and variable construction. Full details of the data construction procedure are provided in Appendix A.

The final panel is organized at the level of country $\times$ tenor $\times$ forecast date $\times$ horizon and contains **3,662 usable forecasting observations** with no missing values in the target variable. The sample covers **16 emerging-market economies** across four geographic regions: Central and Eastern Europe (Czech Republic, Hungary, Poland, Romania, Bulgaria), Southeast Asia (Indonesia, Malaysia, Philippines, Thailand), other Asia (India, South Korea, Taiwan), and Latin America and other emerging markets (Brazil, Mexico, Russia, Turkey). The time series dimension spans **2009 Q1 to 2025 Q2**, a period that encompasses the post-Global Financial Crisis recovery, the 2013 taper tantrum, the 2015–16 commodity price collapse, the 2018 emerging-market stress episode, the COVID-19 shock, and the 2022–23 global monetary tightening cycle. Table 1 reports observation counts and time coverage by country. Several economies—including Romania, Brazil, Russia, Bulgaria, and India—enter the sample later than the full-panel start date, reflecting the availability of consensus forecast coverage in the underlying database.

Table 1: Sample Countries: Observation Counts and Time Coverage

Country	N	Start	End
Hungary	463	2009-Q2	2025-Q2
Czech	428	2009-Q1	2025-Q2
Poland	427	2009-Q1	2025-Q2
Indonesia	413	2009-Q1	2025-Q2
Mexico	250	2009-Q1	2025-Q2
South Korea	232	2009-Q1	2025-Q2
Thailand	229	2009-Q2	2025-Q1
Philippines	225	2010-Q2	2025-Q2
Malaysia	219	2009-Q1	2025-Q2
Romania	190	2017-Q1	2025-Q2
Brazil	162	2017-Q3	2025-Q1
India	132	2015-Q1	2024-Q4
Russia	108	2017-Q3	2025-Q2
Taiwan	97	2015-Q1	2023-Q3
Bulgaria	44	2017-Q3	2021-Q1
Turkey	43	2012-Q3	2025-Q1
Total	3,662	2009-Q1	2025-Q2

3.2 Maturity and Horizon Coverage

The panel covers three maturity buckets—3-month (3M), 2-year (2Y), and 10-year (10Y)—and six forecast horizons: 3, 6, 9, 12, 18, and 24 months. As shown in Table 2, the distribution is severely unbalanced across maturities. The 10Y segment accounts for 2,936 observations (80.2% of the sample), reflecting the broader consensus forecast coverage of long-tenor yields in the underlying database. The 2Y segment contributes 454 observations (12.4%), and the 3M segment the remaining 272 (7.4%). Within each tenor, observation counts decline at longer horizons as fewer forecasts extend to 18 and 24 months, particularly for shorter maturities.

Table 2: Observation Counts by Tenor and Forecast Horizon

Tenor	3M	6M	9M	12M	18M	24M	Total
10Y	600	596	591	578	298	273	2,936
2Y	103	95	85	76	58	37	454
3M	52	51	50	47	37	35	272
Total	755	742	726	701	393	345	3,662

3.3 Target Variable

The main target variable is the professional consensus forecast error:

$$e_{i,m,t,h} = y_{i,m,t+h} - F_{i,m,t,h}^{\text{Con}},$$

where $y_{i,m,t+h}$ is the realized sovereign yield for country i , maturity m , at target date $t + h$, and $F_{i,m,t,h}^{\text{Con}}$ is the professional consensus forecast formed at time t for horizon h . A positive value indicates that the consensus underestimated the realized yield (underprediction); a negative value indicates overprediction. The natural benchmark—predicting $\hat{e} = 0$, i.e., trusting the consensus unconditionally—implies that any model with a positive out-of-sample R^2 adds incremental predictive value beyond the professional consensus.

Table 3 reports summary statistics of the forecast error by tenor. The overall mean is close to zero (0.02 pp), consistent with approximate unconditional unbiasedness on average. However, this masks important heterogeneity: the 3M segment has a notably positive mean error (0.40 pp), while the 10Y segment is near zero (−0.01 pp). Standard deviations range from 1.73 pp (10Y) to 3.15 pp (2Y), and the full range spans −32 pp to +13 pp, reflecting sharp yield movements during the global financial stress episodes in the sample.

Table 3: Summary Statistics of Forecast Error by Tenor (percentage points)

Tenor	N	Mean	Std	Min	P25	Median	P75	Max	<i>Note:</i> Forecast error
10Y	2,936	−0.01	1.73	−25.48	−0.66	−0.12	0.40	12.91	
2Y	454	0.04	3.15	−32.07	−0.37	0.12	0.83	6.88	
3M	272	0.40	2.72	−9.89	−0.62	−0.12	0.70	13.44	
All	3,662	0.02	2.04	−32.07	−0.63	−0.09	0.45	13.44	

defined as realized yield minus professional consensus forecast, in percentage points.

Table 4 reveals a striking pattern across tenors and horizons. At short horizons (3–9 months), the consensus systematically *overpredicts* yields (negative errors) across all three maturity buckets. At longer horizons (18–24 months), the sign reverses and the consensus systematically *underpredicts*. This horizon-dependent bias is consistent across tenors, suggesting a structural feature of professional forecasting behavior. The magnitude of the reversal is largest for the 3M tenor (from −0.48 at 3 months to +2.03 at 24 months), highlighting the potential for machine learning to exploit this systematic pattern.

Table 4: Mean Forecast Error by Tenor and Horizon (percentage points)

Tenor	3M	6M	9M	12M	18M	24M	<i>Note:</i> Positive values indicate consensus
10Y	−0.17	−0.12	−0.11	−0.12	0.37	0.55	
2Y	−0.90	−0.30	−0.14	0.22	1.16	1.79	
3M	−0.48	−0.20	0.03	0.41	1.42	2.03	

underprediction; negative values indicate overprediction.

3.4 Predictors

The predictor set draws on four categories of information, all constructed using only information available at the forecast date t , ensuring strict adherence to the real-time information constraint and eliminating any possibility of look-ahead bias.

Local yield variables: The current local sovereign yield at the forecast date, the consensus forecast level for the target yield, and the forecast revision defined as the change in consensus from the prior quarter. The forecast revision captures the rate at which professional views are being updated.

U.S. yield consensus forecasts: Consensus forecasts of U.S. Treasury yields at the 2-year and 10-year maturities for the same forecast quarter, capturing the expected global interest rate environment at the time the local yield forecast is formed.

Macro consensus forecasts: Consensus projections of local GDP growth, CPI inflation, and the current account balance (all QoQ), as well as corresponding U.S. macro forecasts, all reflecting expectations formed at the forecast date for the same forecast quarter.

Global financial variables: The VIX volatility index, Brent crude oil price, S&P 500 index, gold and silver spot prices, and the bilateral USD/local currency exchange rate, all constructed from weekly observations aggregated to quarterly averages.

Because both the macro predictors and the consensus yield forecasts are drawn from the same consensus forecast database, any incremental predictive value generated by machine learning must reflect more efficient processing of commonly available information—directly paralleling the argument in Bianchi, Ludvigson, and Ma (2022) that machine learning corrects errors in human judgment by better exploiting publicly available data. Summary statistics for the key predictors are reported in Table 5.

Missing value treatment: Eight variables contain missing values: `current_yield_t` (approximately 38% missing), `bbg_consensus_lag1` and `forecast_revision` (5.1% each), `local_gdp_qoq_fcst`, `local_cpi_qoq_fcst`, and `local_ca_qoq_fcst` (16.9% each), `fx_usd_local` (3.6%), and `us_2y_rate` (minor). All three models apply the same imputation strategy: missing values are replaced with the median computed on the training set at each rolling window, and a binary missing indicator is added to the feature set for each affected variable (e.g., `current_yield_t_missing`) to allow the model to learn differential behavior for imputed versus observed values. This strategy is applied consistently across ARX, LightGBM, and LSTM prior to model estimation, ensuring comparability across models. The missing indicator approach is particularly important for `current_yield_t`, whose 38% missingness rate would otherwise substantially reduce the effective sample size.

3.5 Motivating Evidence

Before building machine learning forecasting models, this section formally tests whether consensus forecast errors contain systematic components that are predictable from publicly available information. Table 4 has already shown that the unconditional mean forecast error exhibits a systematic sign reversal across horizons, but unconditional means alone cannot establish whether these patterns are statistically meaningful or whether errors can be predicted by observable information. We

Table 5: Summary Statistics of Key Predictors

Variable	N	Mean	Std	Min	Median	Max
<i>Panel A: Local yield variables</i>						
Current yield at forecast date	2,272	5.05	3.20	−0.02	4.56	31.34
Consensus forecast (local sovereign yield)	3,662	4.92	2.86	0.05	4.55	36.28
Forecast revision (change from prior qtr.)	3,477	0.05	0.81	−7.69	0.00	24.74
<i>Panel B: U.S. yield consensus forecasts (same forecast quarter)</i>						
U.S. 10Y yield consensus forecast	3,662	2.71	1.18	0.66	2.51	4.57
U.S. 2Y yield consensus forecast	3,662	2.09	1.72	0.12	1.62	5.04
<i>Panel C: Macro consensus forecasts (same forecast quarter)</i>						
Local GDP growth (QoQ, %)	3,044	3.16	4.24	−23.13	3.60	22.68
Local CPI (QoQ, %)	3,044	4.09	4.30	− 2.76	3.30	72.28
Local current account (QoQ, %)	3,044	0.73	4.09	− 9.11	−0.04	17.96
U.S. GDP growth (QoQ, %)	3,662	2.38	2.64	− 7.40	2.40	12.40
U.S. CPI (QoQ, %)	3,662	2.96	2.19	− 1.63	2.20	8.67
U.S. current account (QoQ, %)	3,662	−2.75	0.72	−4.54	−2.72	−1.79
<i>Panel D: Global financial variables (weekly, quarterly average)</i>						
VIX index	3,662	18.81	5.83	10.15	16.97	45.13
Brent crude oil (USD/bbl)	3,662	74.77	20.77	33.83	75.08	118.86
S&P 500 index	3,662	3,190	1,257	802	2,958	5,914
FX rate (USD/local currency)	3,530	1,715	4,373	0.81	28.23	16,467

Note: All consensus forecast variables reflect market expectations formed at the forecast date for the same forecast quarter. Global financial variables are constructed from weekly observations aggregated to quarterly averages. `current_yield_t` has $\approx 38\%$ missing values; local macro forecasts have $\approx 17\%$ missing values. FX rate is expressed in local currency units per USD.

address this using the framework of Coibion and Gorodnichenko (2015).

Specification: For each forecast horizon h , we estimate:

$$e_{i,m,t,h} = \alpha + \beta_h \cdot r_{i,m,t,h} + \eta_i + \delta_t + \mu_{i,m,t,h},$$

where $r_{i,m,t,h} = F_t y_{i,m,t+h} - F_{t-1} y_{i,m,t+h}$ is the forecast revision, defined as the difference between the current and prior quarter consensus forecast for the same target date; η_i are country $\times$ tenor fixed effects; δ_t are time fixed effects; and standard errors are clustered at the country $\times$ tenor level. Under rational expectations, $\beta_h = 0$: forecast revisions, as publicly available information at time t , should not predict forecast errors formed at the same date. $\beta_h < 0$ indicates *overreaction*: forecasters tend to overshoot when revising upward. $\beta_h > 0$ indicates *information rigidity*: forecasters underreact to new information.

Results: Table 6 reports estimates for all six forecast horizons. The results display clear heterogeneity across horizons and cannot be summarized by a single behavioral pattern.

At the short horizon (3 months), $\hat{\beta}_3 = -0.894$ ($p < 0.01$), with a within R^2 of 0.436, indicating strong overreaction: forecast revisions explain over 40% of the within-variation in forecast errors

after controlling for fixed effects. The direction is consistent with Table 4: forecasters systematically overshoot when revising short-term yield forecasts upward.

At medium horizons (6 to 12 months), $\hat{\beta}_h$ is statistically indistinguishable from zero ($p > 0.35$) and within R^2 is near zero, suggesting that medium-term forecast errors are not predictable by forecast revisions and are approximately consistent with rational expectations at these horizons.

At long horizons (18 to 24 months), overreaction re-emerges: $\hat{\beta}_{18} = -0.432$ ($p < 0.01$) and $\hat{\beta}_{24} = -0.346$ ($p < 0.01$). Notably, Table 4 shows that the *unconditional mean* forecast error is positive at long horizons (systematic underprediction), while the CG regression reveals *conditional* overreaction with respect to forecast revisions. These two findings are not contradictory: long-horizon forecast errors contain two components—a systematic underprediction bias captured by the unconditional mean, and a conditional overreaction with respect to the direction of forecast revisions captured by the CG regression.

Implications for machine learning: These results reject rational expectations at short and long horizons and confirm that consensus forecast errors contain systematic, predictable components. This directly supports the central premise of this paper: machine learning models can identify and correct systematic biases in professional consensus forecasts. The CG specification uses only a single linear predictor (the forecast revision); the ARX, LightGBM, and LSTM models in this paper augment this with a richer predictor set and allow for non-linear structure, aiming to extract incremental predictive information beyond what the CG specification can capture. If machine learning models outperform the linear ARX, this would imply that forecast errors contain non-linear structure beyond what linear information extraction can recover.

Table 6: Forecast Revision and Forecast Errors: Coibion–Gorodnichenko (2015) Test

	Forecast Horizon					
	3M	6M	9M	12M	18M	24M
$\hat{\beta}_h$	-0.8939*** (0.1328)	-0.0871 (0.1533)	0.1872 (0.2043)	0.2046 (0.2358)	-0.4319*** (0.0900)	-0.3457*** (0.0265)
N	742	726	701	650	370	288
Within R^2	0.436	0.002	0.008	0.006	0.039	0.018

Note: Dependent variable is the consensus forecast error $e_{i,m,t,h}$. The regressor $r_{i,m,t,h}$ is the forecast revision (current minus prior quarter consensus forecast for the same target date). All specifications include country×tenor and time fixed effects. Standard errors in parentheses are clustered at the country×tenor level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

4 Methodology

4.1 Forecasting Framework

The central design of this paper casts the machine learning prediction task as *forecast error correction* rather than direct yield-level prediction. Rather than asking whether machine learning can forecast sovereign yields from scratch, we ask whether it can systematically identify and correct the errors embedded in professional consensus forecasts.

Target variable. The main target variable is the professional consensus forecast error:

$$e_{i,m,t,h} = y_{i,m,t+h} - F_{i,m,t,h}^{\text{Con}},$$

where $y_{i,m,t+h}$ is the realized sovereign yield for country i , maturity m , at target date $t + h$, and $F_{i,m,t,h}^{\text{Con}}$ is the professional consensus forecast formed at forecast date t for horizon h .

Benchmark. The natural benchmark predicts the forecast error as zero:

$$\hat{e}_{i,m,t,h}^{\text{benchmark}} = 0,$$

that is, trusting the consensus unconditionally without adjustment. Machine learning models are therefore held to a high standard: they must demonstrate the ability to systematically identify when the consensus is too optimistic or too pessimistic, and correct for it.

Adjusted yield forecast. The adjusted yield forecast of model k is:

$$\hat{y}_{i,m,t+h}^{(k)} = F_{i,m,t,h}^{\text{Con}} + \hat{e}_{i,m,t,h}^{(k)}.$$

Evaluation metric. The primary accuracy metric is the out-of-sample R^2 (OOS R^2):

$$\text{OOS } R^2 = 1 - \frac{\text{MSE}_{\text{model}}}{\text{MSE}_{\text{consensus}}},$$

where $\text{MSE}_{\text{consensus}}$ is the mean squared forecast error of the consensus benchmark (i.e., $\hat{e} = 0$). A positive OOS R^2 indicates that the model improves upon the consensus; a negative value indicates deterioration. By construction, OOS $R^2 = 0$ corresponds to the consensus benchmark performance. Auxiliary metrics include MSE, RMSE, and MAE.

Connection to Section 3.5. The CG (2015) tests in Section 3.5 confirm that consensus forecast errors contain systematic, predictable components at short (3-month) and long (18- and 24-month) horizons, rejecting rational expectations at these horizons. The forecasting framework in this paper is built directly on this foundation: if forecast errors contain learnable patterns, machine learning models should be able to identify and correct them, generating positive OOS R^2 relative to the consensus benchmark.

4.2 Handling the Unbalanced Panel

The 10Y maturity segment accounts for 80.2% of the total sample, while the 3M segment contributes only 7.4%. This imbalance poses two distinct challenges. First, if models are estimated separately

by maturity, the sparse segments face a severe variance problem: insufficient training observations lead to unstable estimates and degraded out-of-sample performance. Second, if observations are pooled without any correction, the 10Y segment dominates the training process, and the learned forecasting rules effectively reflect 10Y-specific dynamics, undermining any claim that the results are broadly applicable across maturities. This paper adopts a four-layer design to address these challenges.

Layer 1: Tenor and horizon as model features. Tenor (3M, 2Y, 10Y) and forecast horizon (3, 6, 9, 12, 18, 24 months) are included as explicit input features in all models. For ARX, these enter as dummy variables; for LightGBM and LSTM, they are encoded as categorical features, allowing the models to learn non-linear tenor- and horizon-specific effects rather than imposing a common forecasting structure across all maturities and horizons.

Layer 2: Equal weighting. The primary design adopts equal weighting—each observation contributes equally to the training objective, with no additional weight adjustments across maturity segments. This is the most neutral design: although the 10Y segment naturally exerts greater influence during training due to its larger observation count (2,936 observations), each individual 10Y observation carries the same weight as each individual 2Y or 3M observation. The model therefore does not assign additional priority to any maturity segment’s observations beyond what arises from their relative frequency in the sample.

Layer 3: Stratified evaluation. OOS R^2 is reported separately for the full sample, each maturity subsample (3M, 2Y, 10Y), and each forecast horizon. If all three maturity segments exhibit positive OOS R^2 , this supports the generalizability of the results across maturities. If gains are concentrated in the 10Y segment, this would indicate that the machine learning advantage is maturity-specific rather than general.

Layer 4: Inverse frequency weighting robustness check. Inverse frequency weighting is used as a robustness check, with results reported in the Appendix. The weight assigned to each maturity segment is:

$$w_m = \frac{1/N_m}{\sum_{m'} 1/N_{m'}},$$

where N_m is the number of observations in maturity segment m . This scheme assigns higher per-observation influence to sparse segments (2Y, 3M), ensuring that the model does not overfit the 10Y forecasting pattern due to sample imbalance. Comparing equal weighting and inverse frequency weighting results directly reveals the effect of weighting design on maturity-specific performance: inverse frequency weighting improves OOS R^2 for sparse segments but reduces gains for 10Y, since each 10Y observation receives substantially lower relative weight during training.

4.3 Rolling Window Design

Since this paper is a forecasting exercise, model evaluation follows a rolling out-of-sample prediction design rather than random train-test splitting. At each forecast date, the model is estimated using only observations available up to that point, and the fitted model is then used to generate a one-step-ahead prediction of the forecast error. This design strictly replicates the information environment of a real-time forecaster and eliminates any possibility of look-ahead bias.

Window type: The primary design uses an **expanding window**: at each forecast date, all historical observations from the beginning of the sample to that point are used for estimation. The expanding window makes full use of available historical information and is particularly suitable for settings with limited sample size. A fixed rolling window is used as a robustness check.

Minimum training sample threshold: A minimum training sample threshold is imposed to ensure adequate observations for stable estimation at the beginning of the evaluation period. ARX requires relatively few observations for stable OLS estimation; LightGBM and LSTM require larger training sets to learn non-linear structure reliably. Specific thresholds are reported for each model in Section 4.4.

Ablation study – training sequence length on LSTM: As a methodological contribution, we systematically compare short (2 quarters), medium (4 quarters), and long (6 quarters) training sequence configurations and their effect on OOS R^2 , assessing the sensitivity of machine learning performance to the length of the available information horizon and identifying which forecast horizons are most sensitive to sequence length choice.

Statistical significance tests: To confirm that out-of-sample predictive improvements are statistically significant rather than attributable to chance, this paper employs two complementary testing procedures. For *nested model comparisons*—ARX, LightGBM, and LSTM against the consensus benchmark ($\hat{e} = 0$)—the Clark and West (2007) test (CW test) is used. The standard Diebold-Mariano test suffers from size distortion in nested settings; the CW test corrects this by incorporating an adjustment term and is the standard approach in the forecasting literature for testing whether a model with additional predictors improves upon a restricted benchmark. For *non-nested model comparisons*—LightGBM vs. ARX, LSTM vs. ARX, and LSTM vs. LightGBM—the Diebold and Mariano (1995) test (DM test) is applied, using mean squared error as the loss function. All tests use heteroskedasticity-robust standard errors computed at the observation level. Full results are reported in Section 5.7.

4.4 Models

This paper compares one benchmark and three estimated models, forming a progressive comparison structure from linear to non-linear to deep learning.

Consensus benchmark: The consensus benchmark sets $\hat{e} = 0$, trusting the professional consensus unconditionally. All OOS R^2 values are computed relative to this benchmark.

Model 1: ARX (autoregressive model with exogenous variables). ARX extends the standard linear regression model by incorporating a self-referential component alongside exogenous predictors. The specification used in this paper is:

$$e_{i,m,t,h} = \alpha + \beta \cdot r_{i,m,t,h} + \gamma' \mathbf{x}_{i,t} + \delta_m + \delta_h + \varepsilon_{i,m,t,h},$$

where $r_{i,m,t,h}$ is the forecast revision (**forecast revision**), $\mathbf{x}_{i,t}$ is the vector of remaining predictors available at forecast date t , δ_m and δ_h are maturity and horizon fixed effects respectively, and parameters are estimated by OLS.

Why forecast revision rather than lagged forecast errors? The natural autoregressive term

would be the lagged forecast error $e_{i,m,t-1,h}$. However, $e_{i,m,t-1,h}$ requires the realized yield at target date $t - 1 + h$, which has not yet occurred at forecast date t for any horizon beyond 3 months—a direct look-ahead bias. By contrast, the forecast revision $r_{i,m,t,h} = F_t y_{i,m,t+h} - F_{t-1} y_{i,m,t+h}$ is fully observable at forecast date t and captures the same economic intuition: how forecasters are updating their views. Furthermore, the Coibion and Gorodnichenko (2015) regressions in Section 3.5 demonstrate that forecast revisions have significant predictive power for forecast errors at short and long horizons, directly supporting the inclusion of $r_{i,m,t,h}$ in the ARX specification.

ARX serves as a strong linear benchmark, representing the upper bound of what can be achieved by efficiently extracting linear information from the predictor set. If machine learning further outperforms ARX, this implies that the forecast errors contain non-linear structure beyond what linear information extraction can recover.

Model 2: LightGBM (gradient-boosted decision trees). Gradient boosting constructs an ensemble of decision trees iteratively, with each tree fitted to the residuals of the previous ensemble. LightGBM is a high-performance implementation of gradient boosting that adopts a histogram-based splitting algorithm and leaf-wise tree growth strategy, achieving high predictive accuracy at substantially reduced computational cost relative to conventional GBDT implementations. Decision trees naturally capture interaction effects and threshold non-linearities among predictors—structure that is difficult to represent in linear models. LightGBM natively supports the `sample_weight` parameter, making it directly compatible with the equal weighting and inverse frequency weighting schemes described above.

Hyperparameter selection: LightGBM involves multiple hyperparameters that must be selected prior to estimation. This paper adopts time-series cross-validation within each rolling window to select the optimal hyperparameters, avoiding the look-ahead bias that would arise from fixing hyperparameters on the full sample. At each forecast date t , the last 20% of the training set is held out as a validation set; the model is trained on the remaining 80% and evaluated on the validation set by MSE. The search space covers three hyperparameters: learning rate $\in \{0.01, 0.05, 0.10\}$, number of leaves $\in \{15, 31, 63\}$, and number of estimators $\in \{100, 200, 500\}$, yielding 27 candidate combinations. Early stopping is applied throughout: training halts if validation MSE fails to improve for 20 consecutive rounds, preventing overfitting regardless of the nominal `n_estimators` setting. The hyperparameter combination minimizing validation MSE is selected, and the model is re-estimated on the full training set before generating out-of-sample predictions.

LightGBM serves as the primary non-linear model in this paper. If LightGBM further outperforms the linear ARX, this implies that forecast errors contain non-linear structure beyond what linear information extraction can recover—for example, interaction effects between global risk sentiment and domestic fundamentals, or threshold effects specific to certain tenor-horizon combinations. The choice of LightGBM is also supported by strong empirical evidence: in the M5 forecasting competition, all top-50 submissions relied on LightGBM (Makridakis et al., 2022); in the macroeconomic forecasting literature, Goulet Coulombe et al. (2022) show that gradient boosting outperforms sparsity-based models when the predictor space is large and interactions among variables are complex.

Model 3: LSTM (long short-term memory network). LSTM is a variant of recurrent neural networks (RNN) specifically designed to capture long-range temporal dependencies in sequential data. The core mechanism consists of a cell state and three gating structures—input gate, forget gate, and output gate—that allow the network to selectively retain or discard information across time steps, overcoming the vanishing gradient problem that limits standard RNNs.

Input design: Unlike ARX and LightGBM, which take a single observation as input, LSTM takes a sequence of length L as input. For each observation (country i , tenor m , forecast date t , horizon h), the input to the LSTM is the sequence of feature vectors over the past L quarters:

$$\mathbf{X}_{i,m,t,h} = [\mathbf{x}_{i,t-L+1}, \mathbf{x}_{i,t-L+2}, \dots, \mathbf{x}_{i,t}],$$

and the output is the predicted forecast error $\hat{e}_{i,m,t,h}$. The primary specification uses $L = 4$ (four quarters), and ablation experiments with $L = 2$ and $L = 6$ are conducted to assess the sensitivity of model performance to the sequence length choice. Observations for which fewer than L historical periods are available for a given country $\times$ tenor $\times$ horizon combination are excluded from the LSTM evaluation sample; accordingly, the LSTM sample is slightly smaller than that of ARX and LightGBM.

Hyperparameter configuration: Unlike LightGBM, LSTM hyperparameters are held fixed across all rolling windows: hidden units = 64, number of layers = 2, dropout = 0.2, batch size = 32, maximum epochs = 50, learning rate = 0.001 (Adam optimizer), and early stopping patience = 10. This design reflects three considerations. First, the computational cost of LSTM training is substantially higher than gradient boosting; conducting CV-based hyperparameter search across 56 rolling windows would be computationally prohibitive. Second, the approach is consistent with related work: Nunes et al. (2026) adopt a fixed network architecture for their LSTM-LagLasso method without re-tuning at each forecast date, supporting the validity of this design choice. Third, the sequence length ablation study ($L = 2, 4, 6$) serves as an alternative sensitivity analysis for the primary architectural choice.

Role in this paper: LSTM serves as the deep learning extension in this paper. If consensus forecast errors exhibit cross-period persistence—for example, if professional forecasters systematically underestimate the pace of yield increases across multiple consecutive quarters during a global monetary tightening cycle—LSTM can identify these dynamics through its memory mechanism. While ARX and LightGBM can partially capture such patterns through lagged variables, their capacity to represent long-range dependence structures is inherently limited. If LSTM further outperforms LightGBM, this would imply that forecast errors contain temporal dynamic structure beyond static non-linear patterns.

The model set is deliberately kept parsimonious, providing a clean progressive comparison across a strong practical benchmark, a standard econometric model, a flexible non-linear ML model, and a deep learning extension, while avoiding the interpretational difficulties associated with a large model zoo.

5 Results

5.1 ARX Results

The ARX model serves as the linear benchmark, representing the upper bound of what can be achieved by efficiently extracting linear information from the predictor set. A positive OOS R^2 for ARX would confirm that consensus forecast errors contain systematic, linearly predictable components—a finding already suggested by the Coibion and Gorodnichenko (2015) tests in Section 3.5.

Overall performance: ARX achieves an overall OOS R^2 of 0.374, reducing the consensus MSE from 4.395 to 2.752 and RMSE from 2.096 to 1.659. This indicates that the linear model extracts substantial incremental predictive information from the predictor set, reducing mean squared forecast error by 37.4% relative to the professional consensus. Notably, the ARX MAE (1.154) slightly exceeds the consensus MAE (1.085), suggesting that while ARX substantially reduces large errors—which are penalized more heavily under MSE—it introduces some noise in small-error predictions.

Results by maturity: ARX delivers positive OOS R^2 across all three maturity segments, but with meaningful heterogeneity. The 2Y segment performs best (OOS $R^2 = 0.568$), followed by the 3M segment (0.485), while the 10Y segment is the weakest (0.245). This pattern is consistent with the CG (2015) results in Section 3.5: short-maturity forecast errors exhibit stronger linear predictability, while 10Y forecast errors display weaker linear structure, suggesting the presence of non-linear patterns that linear models cannot fully capture.

Results by horizon: The 3M forecast horizon delivers the highest OOS R^2 (0.557), closely mirroring the strong overreaction finding at the 3-month horizon in the CG (2015) tests. The 6M horizon is the weakest (0.272), again consistent with the near-zero CG coefficient at that horizon. Medium to long horizons (9M–24M) show stable performance with OOS R^2 ranging from 0.280 to 0.363.

Tenor $\times$ horizon decomposition: Table 7 reports OOS R^2 across all 18 tenor $\times$ horizon combinations. All 18 cells are positive, confirming that ARX improves upon the consensus across every maturity-horizon combination. The smallest gain appears at 10Y $\times$ 6M (0.098), consistent with the near-zero CG coefficient at the 6-month horizon; the largest gain is at 2Y $\times$ 3M (0.688), indicating that 2Y short-term forecast errors are most amenable to linear correction.

Table 7: ARX: OOS R^2 by Tenor and Forecast Horizon

Tenor	Forecast Horizon					
	3M	6M	9M	12M	18M	24M
10Y	0.376	0.098	0.182	0.210	0.326	0.222
2Y	0.688	0.536	0.557	0.504	0.246	0.172
3M	0.287	0.274	0.530	0.502	0.535	0.493

Note: OOS $R^2 = 1 - \text{MSE}_{\text{model}}/\text{MSE}_{\text{consensus}}$. A positive value indicates improvement over the professional consensus benchmark. Equal weighting is used as the primary specification.

5.2 LightGBM Results

LightGBM serves as the primary non-linear model. A positive incremental gain over ARX would imply that consensus forecast errors contain non-linear structure beyond what linear information extraction can recover.

Overall performance: LightGBM achieves an overall OOS R^2 of 0.490, improving upon ARX by 11.6 percentage points and reducing MSE further from 2.752 to 2.242. Unlike ARX, LightGBM also improves upon the consensus in terms of MAE (0.830 versus 1.085), indicating that the non-linear model reduces both large and small forecast errors. Regarding hyperparameter selection, 60% of rolling windows select `num_leaves` = 15 and 56% select `learning_rate` = 0.10, reflecting a preference for shallow, regularized tree structures consistent with the limited sample size.

Results by maturity: The most notable finding is the large improvement in the 10Y segment: OOS R^2 rises from 0.245 under ARX to 0.480 under LightGBM, nearly doubling the linear model’s gain. This pattern is consistent with the presence of non-linear structure in 10Y forecast errors—including interaction effects between global risk sentiment and domestic fundamentals, and threshold effects under specific market conditions—that the linear ARX cannot capture. The 3M segment also improves modestly (0.485 to 0.519), while the 2Y segment declines slightly (0.568 to 0.496), suggesting that 2Y forecast error dynamics are predominantly linear.

Results by horizon: The most striking contrast with ARX appears in the horizon decomposition. ARX performs best at the 3M horizon (0.557), but LightGBM declines to 0.341 at this horizon. Conversely, LightGBM substantially outperforms ARX at medium to long horizons (9M–24M), with OOS R^2 exceeding 0.525 throughout and reaching 0.620 at the 18M horizon. This pattern suggests that short-horizon forecast errors are predominantly linear—captured well by the forecast revision term in ARX—while medium to long-horizon errors contain non-linear structure that gradient boosting can exploit.

Tenor $\times$ horizon decomposition: Table 8 reports the full tenor $\times$ horizon decomposition. Of 18 cells, 17 are positive; the sole negative value occurs at 3M tenor $\times$ 3M horizon (OOS R^2 = -0.224), where LightGBM underperforms the consensus. This cell represents the shortest-maturity, shortest-horizon combination, where high idiosyncratic noise likely causes the non-linear model to overfit. The strongest performance appears at 2Y $\times$ 18M (0.710), suggesting that 2Y medium-term forecast errors contain the richest learnable non-linear dynamics.

Table 8: LightGBM: OOS R^2 by Tenor and Forecast Horizon

Tenor	Forecast Horizon					
	3M	6M	9M	12M	18M	24M
10Y	0.358	0.408	0.467	0.498	0.577	0.511
2Y	0.358	0.603	0.669	0.538	0.710	0.604
3M	-0.224	0.058	0.457	0.621	0.703	0.556

Note: OOS R^2 = $1 - \text{MSE}_{\text{model}} / \text{MSE}_{\text{consensus}}$. A positive value indicates improvement over the professional consensus benchmark. Equal weighting is used as the primary specification. Hyperparameters are selected via time-series cross-validation within each rolling window.

5.3 Deep Dive on Feature Importance: SHAP

To further understand the sources of LightGBM’s predictive gains, this section applies SHAP (SHapley Additive exPlanations) values to decompose each prediction into the marginal contributions of individual features. SHAP values are grounded in cooperative game theory and provide an additive, theoretically consistent decomposition of model predictions. Unlike conventional feature importance measures (e.g., split counts or information gain), SHAP values reveal not only the average magnitude of each feature’s contribution but also its *directional* effect—whether high or low feature values push the predicted forecast error upward or downward. The SHAP analysis is based on a LightGBM model re-estimated on the full training sample (2009 Q1 to 2025 Q1) using the optimal hyperparameters from the final rolling window.

It should be noted that this SHAP analysis captures average feature importance over the entire sample period. Feature importance may vary over time—for example, the relative importance of U.S. rate expectations may have increased substantially during the 2022–23 tightening cycle relative to the pre-2022 period. A rolling SHAP analysis across multiple time points would be a natural extension for future research.

Overall feature importance: Figure 1 presents the SHAP summary plot alongside the mean absolute SHAP values for the top 15 features across the full sample.

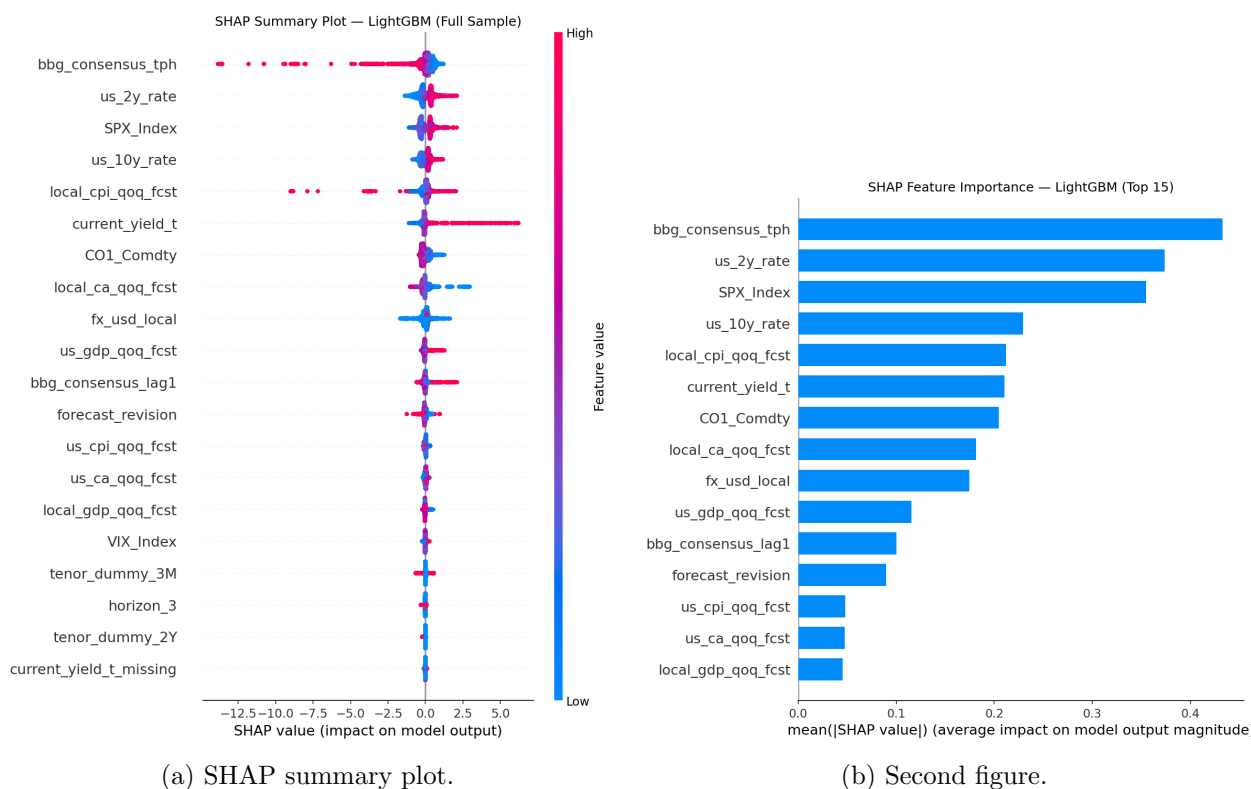


Figure 1: LightGBM SHAP Analysis — Full Sample. Panel (a) shows the SHAP summary plot, where each point represents one observation, the x-axis shows the SHAP value (impact on model output), and color indicates the feature value (red = high, blue = low). Panel (b) shows mean absolute SHAP values for the top 15 features.

bbg_consensus_tph (the consensus forecast level) is the single most important feature (mean $|\text{SHAP}| = 0.44$). This result warrants careful interpretation. **bbg_consensus_tph** is the consensus forecast of the sovereign yield for target date $t + h$, formed at forecast date t and therefore fully observable at the time of prediction. The model learns from historical data that the level of the consensus forecast is systematically informative about the direction of the subsequent forecast error: when the consensus forecast is high, the realized yield has historically tended to fall *below* the consensus (negative error); and vice versa. This pattern is clearly visible in the SHAP summary plot—red points (high consensus forecast values) cluster in the negative region, while blue points (low consensus forecast values) cluster in the positive region—reflecting the systematic over-reaction documented by Coibion and Gorodnichenko (2015), with the additional finding that this relationship exhibits non-linear threshold structure that the linear ARX cannot fully capture. The appearance of **bbg_consensus_tph** as a feature is a natural property of the forecast error correction framework: the model is not predicting yield levels, but rather learning to identify and correct the systematic directional biases embedded in the consensus.

us_2y_rate (U.S. 2-year yield consensus forecast) ranks second (mean $|\text{SHAP}| = 0.38$), with a clear directional effect: high U.S. short-end rate expectations are associated with positive forecast errors (consensus underprediction) for emerging-market sovereign yields. This is consistent with Jaramillo and Weber (2013), who document that U.S. interest rate shocks are among the primary drivers of emerging-market yield dynamics, and suggests that U.S. monetary policy expectations generate systematic forecast errors that LightGBM can exploit.

SPX_Index (S&P 500 index) ranks third (mean $|\text{SHAP}| = 0.35$), reflecting the non-linear impact of global risk sentiment on emerging-market yield forecast errors. The directional effect is not uniform—equity market movements interact with other financial conditions in complex ways—consistent with the non-linear transmission mechanism documented in International Monetary Fund (2025).

forecast_revision ranks 12th (mean $|\text{SHAP}| = 0.09$), substantially lower than its role as the primary autoregressive term in ARX. In the richer feature environment of LightGBM, the marginal contribution of the forecast revision is diluted by other predictors—particularly **bbg_consensus_tph** and U.S. rate factors—suggesting that non-linear models can extract predictive information from a broader information set beyond the revision signal alone.

Feature importance by maturity: Figure 2 reports SHAP bar plots separately for each maturity segment.

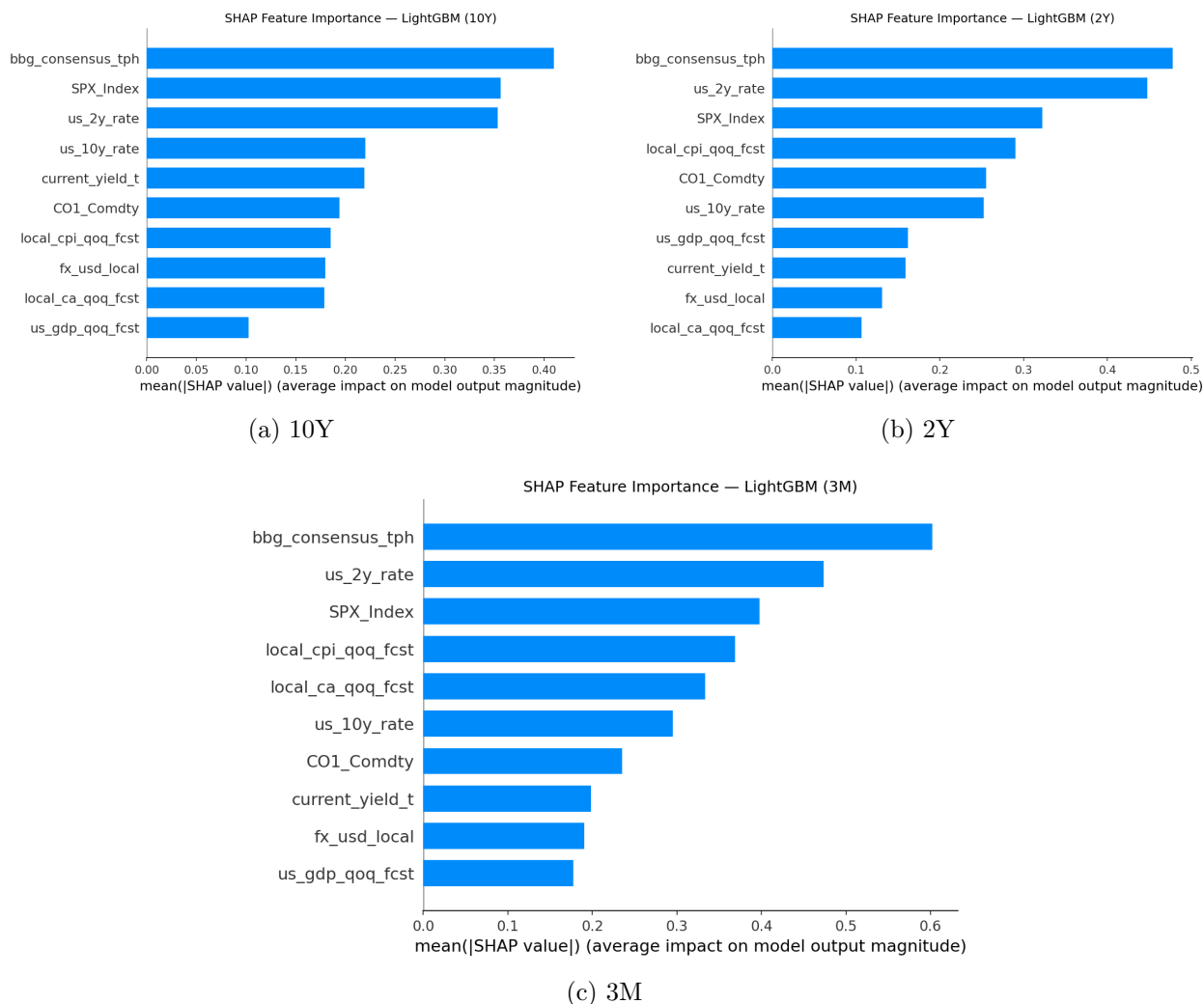


Figure 2: LightGBM SHAP Feature Importance by Tenor (Top 10). Bars show mean absolute SHAP values.

bbg_consensus_tph is the top-ranked feature across all three maturity segments, confirming the universality of the consensus overreaction pattern. However, the composition of the remaining top features differs meaningfully across maturities.

For the **10Y segment**, **SPX_Index** ranks second—above **us_2y_rate**—indicating that long-tenor sovereign yields are more sensitive to global equity market sentiment than to short-end U.S. rate expectations. In addition, **current_yield_t** enters the top five, suggesting a significant mean-reversion dynamic in long-tenor yield forecast errors: when current yields are high, the consensus tends to overestimate future levels, and vice versa.

For the **2Y segment**, **local_cpi_qoq_fcst** (local CPI consensus forecast) ranks fourth—higher than in the 10Y segment—consistent with the greater sensitivity of 2-year sovereign yields to do-

mestic inflation expectations through the central bank policy reaction function. `C01_Comdty` (Brent crude oil) also enters the top five, likely reflecting the commodity price-to-inflation transmission channel.

For the **3M segment**, `local_ca_qoq_fcst` (local current account consensus forecast) enters the top five, reflecting the higher sensitivity of short-maturity yields to external imbalances and short-term capital flow dynamics. Overall, the 3M segment displays the most evenly distributed feature importance profile, with no single feature dominating, suggesting that short-tenor forecast errors are driven by a combination of multiple factors rather than a single dominant channel.

Summary: The SHAP analysis identifies three primary sources of LightGBM’s predictive gains: (1) the non-linear overreaction effect embedded in the consensus forecast level; (2) the non-linear transmission of U.S. interest rate expectations to emerging-market yield forecast errors; and (3) complex interaction effects between global risk sentiment and domestic fundamentals. The maturity-specific heterogeneity in feature importance explains why LightGBM achieves the largest improvement over ARX in the 10Y segment: long-tenor yield forecast errors are driven by the interaction between equity market sentiment and consensus overreaction—precisely the non-linear structure that gradient boosting is designed to capture.

5.4 LSTM Results

This section reports the out-of-sample forecasting performance of the LSTM model, which in total costs two-days to train. LSTM serves as the deep learning extension, with its core advantage lying in the ability to capture cross-period temporal dynamics in forecast errors through its memory mechanism: if professional forecasters make systematic errors in the same direction across multiple consecutive quarters, LSTM can identify and correct this pattern. Unlike ARX and LightGBM, which take a single observation as input, LSTM uses a sequence of length $L = 4$. The evaluation sample is therefore slightly smaller ($N = 2,973$, compared to 3,437 for ARX and LightGBM), as observations for country $\times$ tenor $\times$ horizon combinations with fewer than four historical periods are excluded.

Overall performance: LSTM achieves an overall OOS R^2 of 0.470, falling between ARX (0.374) and LightGBM (0.490). LSTM’s MAE (0.731) is the lowest among all three models, indicating superior performance in reducing small forecast errors. It is important to note that LSTM’s MSE (1.932) cannot be directly compared to LightGBM’s MSE (2.242) due to the smaller evaluation sample; OOS R^2 is the appropriate cross-model comparison metric, as it is computed relative to the consensus benchmark within each model’s own sample.

Results by maturity: LSTM exhibits a strikingly different maturity profile from ARX and LightGBM. The 2Y segment improves substantially to OOS $R^2 = 0.794$, far exceeding LightGBM’s 0.496, implying that 2Y sovereign yield forecast errors contain strong cross-period temporal dynamics: professional forecasters’ systematic biases in 2Y yields persist across consecutive quarters, and LSTM’s memory mechanism effectively identifies and corrects this pattern. The 3M segment also improves markedly (0.519 to 0.733), suggesting that short-maturity forecast errors also exhibit rich temporal structure. By contrast, the 10Y segment declines from LightGBM’s 0.480 to 0.349, indicating that 10Y forecast errors are driven by static non-linear structure rather than cross-period

dynamics, where gradient boosting retains the advantage.

Results by horizon: The contrast between LSTM and LightGBM is sharpest in the horizon decomposition. At the 3-month horizon, OOS R^2 declines further from LightGBM’s 0.341 to 0.191, suggesting that very short-horizon forecast errors lack cross-period persistence and that the temporal memory mechanism introduces noise rather than signal at this horizon. At the 18-month horizon, however, OOS R^2 reaches 0.706, surpassing LightGBM’s 0.620 and becoming the highest value among all three models at this horizon. This directly validates the motivation for including LSTM: during sustained shocks such as global monetary tightening cycles, long-horizon forecast errors exhibit persistent cross-quarter patterns that the LSTM memory mechanism can exploit.

Tenor $\times$ horizon decomposition: Table 9 reports OOS R^2 across all 18 tenor $\times$ horizon combinations. Two cells are negative: 10Y $\times$ 3M (-0.316) and 3M tenor $\times$ 3M horizon (0.123, near zero). For the 10Y short-horizon cell, the absence of temporal persistence means that LSTM’s sequential input introduces noise, underperforming the static LightGBM. The strongest cell is 2Y $\times$ 3M (0.919), indicating that 2Y short-horizon forecast errors have exceptionally strong cross-period memory structure: once the consensus begins to systematically underestimate 2Y yields, this bias tends to persist across multiple quarters.

Table 9: LSTM ($L = 4$): OOS R^2 by Tenor and Forecast Horizon

Tenor	Forecast Horizon					
	3M	6M	9M	12M	18M	24M
10Y	-0.316	0.310	0.452	0.450	0.661	0.435
2Y	0.919	0.202	0.480	0.526	0.789	0.752
3M	0.123	0.580	0.705	0.715	0.803	0.732

Note: OOS $R^2 = 1 - \text{MSE}_{\text{model}}/\text{MSE}_{\text{consensus}}$. A positive value indicates improvement over the professional consensus benchmark. Equal weighting and sequence length $L = 4$ are used as the primary specification.

Observations with fewer than L historical periods available are excluded, yielding $N = 2,973$.

Ablation study: sequence length: Table 10 reports OOS R^2 for sequence lengths $L = 2, 4$, and 6. Overall performance is similar across all three specifications (0.470–0.483), confirming that LSTM results are robust to the choice of sequence length. $L = 2$ achieves the highest overall OOS R^2 (0.483), while $L = 4$ performs best in the 2Y segment (0.794 versus 0.534 for $L = 2$), reflecting the need for a longer memory window to capture 2Y yield temporal dynamics. $L = 6$ performs between the two (0.475 overall) and does not improve upon $L = 4$ in any maturity segment, suggesting that extending the sequence beyond four periods yields no additional gains. This paper adopts $L = 4$ as the primary specification, achieving the best balance between overall robustness and 2Y segment performance.

Table 10: LSTM Ablation Study: OOS R^2 by Sequence Length

Sequence Length	Overall	10Y	2Y	3M
$L = 2$	0.483	0.404	0.534	0.759
$L = 4$	0.470	0.349	0.794	0.733
$L = 6$	0.475	0.416	0.516	0.719

Note: Equal weighting. Bold indicates the highest value in each column. OOS R^2 is computed on each specification’s own evaluation sample.

5.5 Model Overall Comparison and Summary

In this subsection, we integrate the out-of-sample forecasting performance of all three models, provide a systematic cross-model comparison, and summarize the core empirical findings of this paper. Bootstrap confidence intervals for all OOS R^2 estimates are reported in Appendix C.

Overall performance—substantial improvement over the consensus benchmark: Table 11 reports OOS R^2 , MSE, RMSE, and MAE for all four specifications (consensus benchmark, ARX, LightGBM, LSTM) across the full sample, each maturity segment, and each forecast horizon. All three estimated models outperform the consensus benchmark overall, forming a clear progressive structure: ARX (0.374) $\rightarrow$ LSTM (0.470) $\rightarrow$ LightGBM (0.490). This progression reflects three layers of predictive information. First, linear information extraction (ARX) already captures substantial incremental predictive ability from publicly available predictors, supporting the CG (2015) diagnostic results in Section 3.5. Second, static non-linear structure (LightGBM) delivers further improvement, indicating that interaction effects and threshold non-linearities among predictors are present but inaccessible to linear models. Third, temporal dynamics (LSTM) performs comparably to LightGBM overall but exhibits unique advantages in specific maturity-horizon combinations.

Results by maturity: The three models exhibit distinct maturity-specific advantage profiles. In the 10Y segment, LightGBM performs best (0.480), while LSTM is weakest (0.349), indicating that 10Y forecast errors are driven by static non-linear structure. In the 2Y segment, LSTM dominates (0.794), substantially outperforming both LightGBM (0.496) and ARX (0.568), implying rich cross-period temporal dynamics in 2Y yield forecast errors. In the 3M segment, LSTM also outperforms LightGBM (0.733 vs. 0.519), with ARX in between (0.485). This pattern reveals an important economic implication: forecast errors across different maturities are governed by different types of structure, and no single model dominates across all maturities.

Results by horizon: Cross-model differences are equally systematic in the horizon dimension. At the 3-month horizon, ARX performs best (0.557), LightGBM second (0.341), and LSTM weakest (0.191): short-horizon forecast errors follow predominantly linear patterns, with the forecast revision term in ARX sufficient to capture the dominant signal, while temporal memory introduces noise. At medium to long horizons (9M–24M), both LightGBM and LSTM outperform ARX, with their relative advantage growing with the horizon. At the 18-month horizon, LSTM reaches its peak (0.706), surpassing LightGBM (0.620), consistent with the hypothesis that sustained global shocks leave cross-quarter persistent patterns in long-horizon forecast errors that the LSTM memory mechanism can exploit.

Table 11: Model Comparison: Out-of-Sample Forecasting Performance

	Consensus	ARX	LightGBM	LSTM
<i>Panel A: Overall</i>				
OOS R^2	0.000	0.374	0.490	0.470
MSE	4.395	2.752	2.242	—
RMSE	2.096	1.659	1.497	—
MAE	1.085	1.154	0.830	0.731
N	3,437	3,437	3,437	2,973
<i>Panel B: OOS R^2 by Tenor</i>				
10Y	0.000	0.245	0.480	0.349
2Y	0.000	0.568	0.496	0.794
3M	0.000	0.485	0.519	0.733
<i>Panel C: OOS R^2 by Horizon</i>				
3M	0.000	0.557	0.341	0.191
6M	0.000	0.272	0.449	0.321
9M	0.000	0.342	0.529	0.481
12M	0.000	0.325	0.525	0.496
18M	0.000	0.363	0.620	0.706
24M	0.000	0.280	0.529	0.528

Note: OOS $R^2 = 1 - \text{MSE}_{\text{model}}/\text{MSE}_{\text{consensus}}$. Bold indicates the best-performing estimated model in each row. MSE and RMSE for LSTM are not directly comparable to ARX and LightGBM due to the smaller evaluation sample ($N = 2,973$); OOS R^2 is the appropriate cross-model comparison metric. The Clark-West and Diebold-Mariano tests reported in Section 5.7 are conducted on the common evaluation sample ($N = 2,973$), confirming that relative performance rankings are robust to sample composition.

Equal weighting is used as the primary specification for all models.

Tenor $\times$ horizon cross-model comparison: Tables 7, 8, and 9 report the tenor $\times$ horizon OOS R^2 decomposition for all three models. Three features are worth highlighting. First, all three models produce their weakest or negative results at the 3M tenor $\times$ 3M horizon cell, confirming that the shortest-maturity, shortest-horizon combination is the most difficult forecasting problem in this panel, where the consensus benchmark is hardest to beat. Second, 2Y $\times$ 3M is LSTM’s strongest cell (0.919), while ARX also performs well there (0.688), confirming that 2Y short-horizon forecast error predictability is robust across model classes, but LSTM captures an additional temporal layer that linear models cannot represent. Third, LightGBM achieves its strongest results at 10Y long-horizon cells (18M: 0.577, 24M: 0.511), suggesting that long-horizon 10Y forecast errors are governed by non-linear interactions among static macro-financial predictors.

Summary of core findings: Integrating the results above, the core empirical findings of this paper can be summarized in four points.

Finding 1. Consensus forecast errors contain systematic, learnable patterns. All three models outperform the consensus overall, with OOS R^2 ranging from 0.374 to 0.490, corroborating the CG (2015) diagnostic results in Section 3.5: rational expectations are rejected at both short and long horizons.

Finding 2. Non-linear structure delivers incremental predictive value. LightGBM nearly doubles ARX’s OOS R^2 in the 10Y segment (0.245 to 0.480); SHAP analysis attributes this primarily to non-linear interaction effects among the consensus forecast level, U.S. interest rate factors, and global equity market sentiment.

Finding 3. Temporal dynamics are critical in specific contexts. LSTM achieves the strongest performance in the 2Y segment (0.794) and at the 18-month horizon (0.706), indicating that 2Y yield forecast errors and long-horizon errors exhibit cross-quarter persistence that machine learning can exploit through its memory mechanism.

Finding 4. No single model dominates across all dimensions. ARX is strongest at the 3M horizon, LightGBM at the 10Y maturity, and LSTM at the 2Y maturity and long horizons. This suggests that in practical applications, the optimal model choice may vary by target maturity and forecast horizon.

To assess economic significance, we translate RMSE improvements into basis points. The consensus benchmark produces an average forecast error of 209.6 basis points (RMSE = 2.096%). ARX reduces this to 165.9 basis points, a saving of approximately 44 basis points per forecast. LightGBM further reduces the average error to 149.7 basis points, saving approximately 60 basis points relative to the consensus. For sovereign yield investors and policy analysts, a systematic reduction of 44–60 basis points in average forecast error represents economically meaningful improvements in position sizing, risk management, and policy assessment.

5.6 Crisis vs. Non-Crisis Regimes

This section examines whether the predictive gains of machine learning vary with the global financial stress environment. Bianchi et al. (2022) argue that professional forecasters are most prone to systematic errors when market conditions shift abruptly; Jaramillo and Weber (2013) document state-dependent transmission of global risk sentiment to emerging-market sovereign yields. This paper therefore hypothesizes that the incremental predictive value of machine learning is largest in high-stress regimes.

Regime classification: The quarterly average of the VIX (CBOE Volatility Index) is used as a proxy for global financial stress. VIX is widely employed in the literature to proxy for global investor risk aversion and global financial conditions, particularly in emerging-market research (Csontó and Ivaschenko, 2013; International Monetary Fund, 2025). Since the crisis dummy variables in this paper’s dataset (sovereign debt crisis, currency crisis, banking crisis) are never triggered during the sample period, VIX quantiles provide a continuous, real-time observable alternative stress indicator. Two thresholds are used. The **primary definition** classifies quarters as *stress regimes* when the quarterly average VIX exceeds its sample median (17.14), and as *non-stress regimes* otherwise. The **robustness definition** uses the 75th percentile (23.35) to define *crisis regimes*, corresponding to the top quartile of global financial stress in the sample, which encompasses the post-GFC recovery period, the 2011–12 European sovereign debt crisis, the COVID-19 shock of 2020, and the 2022 global monetary tightening cycle. Figure 4 presents the time series of VIX and the resulting regime classification under both thresholds.

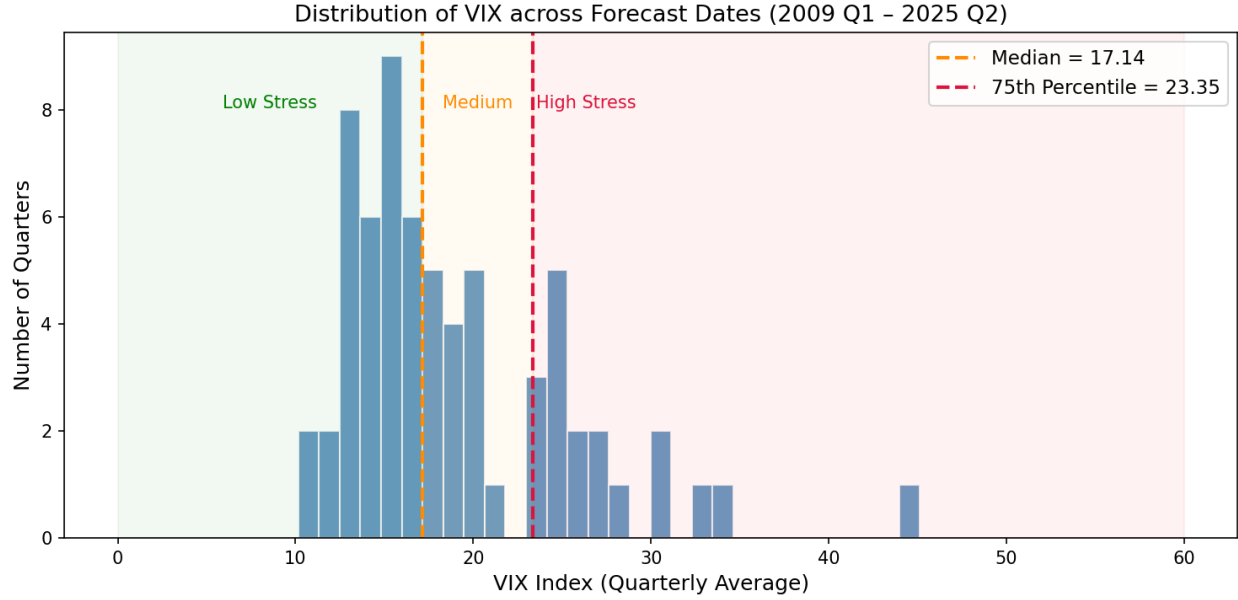


Figure 3: VIX Index Distribution

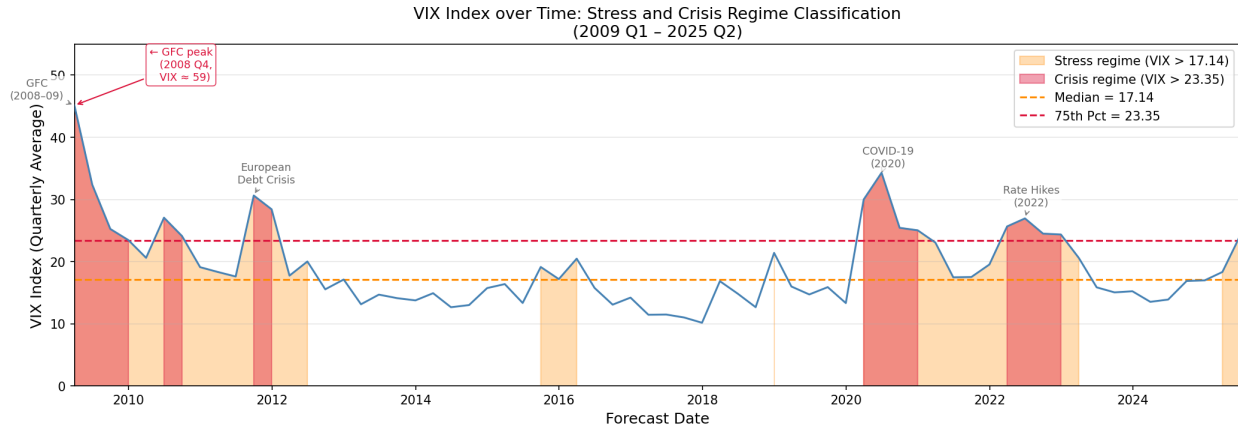


Figure 4: VIX Index over Time: Stress and Crisis Regime Classification (2009 Q1–2025 Q2). Orange shading indicates stress regimes ($VIX > 17.14$); red shading indicates crisis regimes ($VIX > 23.35$). Dashed lines mark the median and 75th percentile thresholds. The GFC peak (2008 Q4, $VIX \approx 59$) falls outside the sample period.

Main results (median threshold): Table 12 reports OOS R^2 for all three models under stress and non-stress regimes. The results display a clear and economically interpretable pattern. ARX performs better in the *non-stress regime* (0.432 vs. 0.311), consistent with the view that linear patterns are more stable when global financial conditions are calm and the forecast revision term efficiently captures short-horizon linear biases. LightGBM performs substantially better in the *stress regime* (0.548 vs. 0.435), an improvement of 11.3 percentage points, directly supporting the paper’s central hypothesis: during periods of global financial stress, non-linear interaction effects among predictors—such as the interaction between VIX and the consensus forecast level—

are more pronounced, and gradient boosting can effectively exploit these patterns. LSTM also performs better in the stress regime (0.532 vs. 0.363), with a gap of 16.9 percentage points—the largest regime difference among all three models, indicating that forecast errors exhibit stronger cross-period temporal persistence during high-stress episodes, where LSTM’s memory mechanism is most valuable.

Table 12: OOS R^2 by Stress Regime

	ARX		LightGBM		LSTM	
	Non-stress	Stress	Non-stress	Stress	Non-stress	Stress
<i>Panel A: Median threshold (VIX = 17.14)</i>						
Overall	0.432	0.311	0.435	0.548	0.363	0.532
10Y	0.309	0.167	0.479	0.480	0.367	0.331
2Y	0.631	0.473	0.359	0.702	0.173	0.810
3M	0.524	0.475	0.435	0.540	0.179	0.751
<i>Panel B: P75 threshold (VIX = 23.35)</i>						
Overall	0.445	0.048	0.517	0.367	0.545	0.208
10Y	0.316	−0.039	0.512	0.351	0.416	0.116
2Y	0.621	0.088	0.497	0.485	0.863	0.287
3M	0.558	0.286	0.592	0.320	0.859	0.473

Note: OOS $R^2 = 1 - \text{MSE}_{\text{model}}/\text{MSE}_{\text{consensus}}$. Bold indicates the better-performing regime for each model-maturity combination. Stress regime: VIX > median (17.14). Crisis regime: VIX > 75th percentile (23.35).

Maturity and horizon heterogeneity: The stress-regime gains are not uniformly distributed. In the maturity dimension, LightGBM’s 2Y gain is the most pronounced (non-stress: 0.359 vs. stress: 0.702), suggesting that 2Y yield forecast errors have the richest non-linear structure during high-stress episodes. LSTM’s 2Y segment shows an equally dramatic stress-regime improvement (non-stress: 0.173 vs. stress: 0.810). In the horizon dimension, LSTM’s 3M horizon exhibits the most striking reversal: OOS $R^2 = -0.431$ in the non-stress regime (worse than the consensus), but 0.742 in the stress regime. This indicates that LSTM’s short-horizon temporal memory introduces noise in calm periods but precisely captures the cross-period dynamics of panic propagation during high-stress episodes.

Robustness results (Applying P75 threshold): Under the stricter crisis definition (VIX > 23.35), an interesting reversal emerges: all three models deteriorate substantially in extreme crisis conditions. ARX nearly vanishes (0.048), while LightGBM (0.367) and LSTM (0.208) also decline sharply. This suggests that during episodes of extreme global stress—primarily the COVID-19 shock—consensus forecasts themselves become highly unstable, reducing the regularity of forecast errors and making systematic outperformance difficult for all models. This finding is consistent with the “prestakes” concept of Bianchi et al. (2025): even machine learning cannot systematically outperform the benchmark during the most extreme episodes of market turbulence.

Summary: Combining the results across both thresholds, machine learning’s predictive advantage is most pronounced in *moderate stress regimes* (VIX between approximately 17 and 23), where

global financial condition non-linearities are most active but have not yet reached extreme chaos. In tranquil periods, the linear ARX is relatively effective; in extreme crisis periods, all models face the challenge of consensus forecasts themselves becoming unreliable. This “sweet spot” finding has practical implications: systematic application of machine learning-assisted forecast correction adds the most value when global financial stress is moderately elevated but not yet out of control.

5.7 Statistical Significance Tests

This section formally tests whether the out-of-sample predictive gains documented in Sections 5.1–5.4 are statistically significant, or could plausibly arise by chance.

Testing methodology: Two complementary tests are employed. For *nested model comparisons*—ARX, LightGBM, and LSTM against the consensus benchmark ($\hat{e} = 0$)—this paper applies the Clark and West (2007) test (CW test). The standard Diebold-Mariano test suffers from size distortion in nested settings; the CW test corrects this by incorporating an adjustment term and is the standard approach in the forecasting literature for testing whether a model with additional predictors improves upon a restricted benchmark. For *non-nested model comparisons*—LightGBM vs. ARX, LSTM vs. ARX, and LSTM vs. LightGBM—this paper applies the Diebold and Mariano (1995) test (DM test), using mean squared error as the loss function. All tests use heteroskedasticity-robust standard errors computed at the observation level.

Clark-West test results (vs. consensus benchmark): Table 13 reports CW test statistics and p-values. All three models reject the null hypothesis that the consensus benchmark performs no worse than the estimated model at the 1% level in the full sample (CW statistics: ARX = 9.32, LightGBM = 10.82, LSTM = 8.60), confirming that the OOS R^2 improvements documented above are statistically significant and not attributable to chance. Results are similarly robust across maturity and horizon subsamples: of the 18 maturity $\times$ horizon subgroups, the vast majority achieve significance at the 5% or 1% level, with a small number reaching 10% significance in smaller subsamples (e.g., the 2Y segment, $N = 317$).

Diebold-Mariano test results (between models): Table 14 reports DM test statistics and p-values for pairwise model comparisons. Three patterns are noteworthy.

LightGBM significantly outperforms ARX. In the full sample (DM = -6.33 , $p < 0.01$) and across almost all maturity and horizon subsamples, LightGBM’s predictive accuracy is statistically significantly higher than ARX’s. The sole exception is the 3M maturity segment (DM = -0.40 , $p = 0.69$), indicating that at the shortest maturity, the linear model already captures the learnable component of forecast errors as effectively as gradient boosting.

LSTM significantly outperforms ARX overall, with concentrated gains. In the full sample, LSTM significantly outperforms ARX (DM = -2.39 , $p < 0.05$), but the improvement is concentrated in the 2Y segment (DM = -3.13 , $p < 0.01$) and 3M maturity (DM = -2.86 , $p < 0.01$). In the 10Y segment, LSTM’s improvement over ARX is not statistically significant ($p = 0.23$), consistent with the finding that temporal dynamics add limited value for long-tenor yield forecast errors.

LSTM and LightGBM are statistically indistinguishable overall. Despite their similar OOS R^2 point estimates (0.470 vs. 0.490), the DM test confirms that the overall difference between LSTM and LightGBM is not statistically significant (DM = 0.18 , $p = 0.86$). The sole exception is the

3M maturity segment, where LSTM significantly outperforms LightGBM ($DM = -3.57$, $p < 0.01$), confirming that short-maturity yield forecast errors contain temporal dynamics that are statistically significant beyond what static non-linear models can capture.

Table 13: Clark-West (2007) Test: Models vs. Consensus Benchmark

	ARX			LightGBM			LSTM		
	Stat		p	Stat		p	Stat		p
<i>Panel A: By Maturity</i>									
Overall	9.32	0.000	***	10.82	0.000	***	8.60	0.000	***
10Y	9.94	0.000	***	14.11	0.000	***	11.32	0.000	***
2Y	2.29	0.011	**	2.34	0.010	***	1.93	0.027	**
3M	5.02	0.000	***	5.01	0.000	***	4.36	0.000	***
<i>Panel B: By Horizon</i>									
3M	2.10	0.018	**	2.01	0.022	**	1.46	0.073	*
6M	6.83	0.000	***	6.75	0.000	***	4.69	0.000	***
9M	8.44	0.000	***	7.19	0.000	***	5.76	0.000	***
12M	9.04	0.000	***	7.42	0.000	***	6.16	0.000	***
18M	6.79	0.000	***	7.45	0.000	***	6.12	0.000	***
24M	8.11	0.000	***	7.36	0.000	***	5.86	0.000	***

Note: One-sided CW test. H_0 : consensus benchmark performs no worse than the estimated model. $*p < 0.10$, $**p < 0.05$, $***p < 0.01$. Evaluation sample: $N = 2,973$ (full sample, common to all three models).

Table 14: Diebold-Mariano (1995) Test: Pairwise Model Comparisons

	LightGBM vs ARX			LSTM vs ARX			LSTM vs LightGBM		
	Stat	p		Stat	p		Stat	p	
<i>Panel A: By Maturity</i>									
Overall	-6.33	0.000	***	-2.39	0.017	**	0.18	0.861	
10Y	-5.58	0.000	***	-1.21	0.227		1.07	0.283	
2Y	-6.18	0.000	***	-3.13	0.002	***	-1.54	0.124	
3M	-0.40	0.691		-2.86	0.004	***	-3.57	0.000	***
<i>Panel B: By Horizon</i>									
3M	-0.02	0.981		0.66	0.511		0.79	0.428	
6M	-4.80	0.000	***	-2.37	0.018	**	-0.48	0.629	
9M	-3.85	0.000	***	-2.61	0.009	***	-0.50	0.616	
12M	-4.55	0.000	***	-2.25	0.024	**	0.22	0.825	
18M	-4.27	0.000	***	-3.34	0.001	***	-0.97	0.333	
24M	-4.44	0.000	***	-2.06	0.040	**	-0.10	0.917	

Note: Two-sided DM test with MSE loss function. Negative DM statistic indicates the first-named model has lower MSE. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Evaluation sample: $N = 2,973$.

Summary: The statistical significance tests comprehensively suggest the core empirical findings of this paper. All three models statistically significantly outperform the consensus benchmark; LightGBM statistically significantly outperforms ARX in most settings; and LSTM and LightGBM are statistically indistinguishable overall, but LSTM’s temporal memory mechanism delivers statistically significant additional gains in the 3M maturity segment.

5.8 Consensus-Level Benchmark

A natural concern is whether the predictive gains documented in Sections 5.1–5.4 primarily reflect the linear relationship between the consensus forecast level (`bbg_consensus_tph`) and subsequent forecast errors, rather than the broader feature set or non-linear interactions exploited by the full models. To address this, we estimate a Consensus-Level Benchmark (CLB) that uses only `bbg_consensus_tph` as a single predictor in a linear rolling regression:

$$\hat{e}_{i,m,t,h} = \alpha + \beta \cdot F_{i,m,t,h}^{\text{Con}} + \varepsilon$$

The CLB is estimated under the same expanding window design as the full models, with the same minimum training size requirement ($N_{\min} = 200$).

Results. Figure 5 and Figure 6 report the CLB OOS R^2 alongside the full models, for the common evaluation sample ($N = 2,973$) and the full sample ($N = 3,437$) respectively.

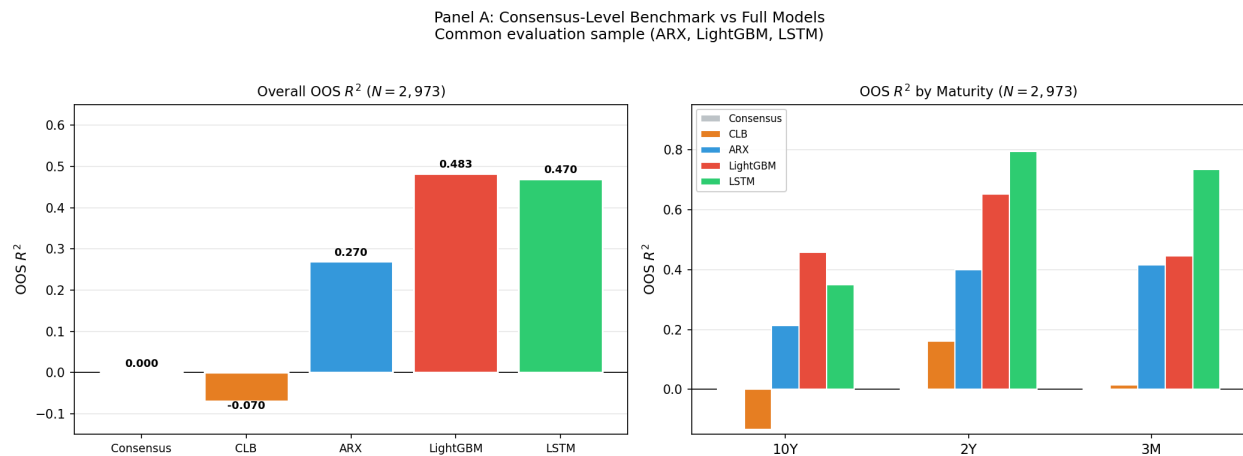


Figure 5: Consensus-Level Benchmark vs. Full Models: Common Evaluation Sample ($N = 2,973$). Panel (a) shows overall OOS R^2 ; Panel (b) shows OOS R^2 by maturity segment. CLB uses only `bbg_consensus_tph` as a single linear predictor. The common sample includes all observations for which ARX, LightGBM, and LSTM predictions are jointly available.

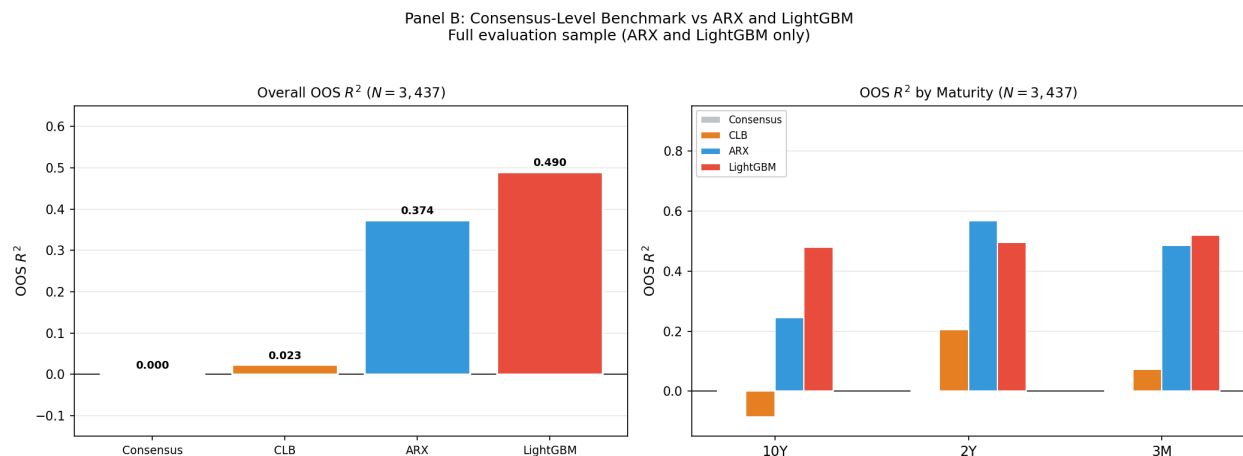


Figure 6: Consensus-Level Benchmark vs. ARX and LightGBM: Full Sample ($N = 3,437$). Panel (a) shows overall OOS R^2 ; Panel (b) shows OOS R^2 by maturity segment. LSTM is excluded from this panel as its evaluation sample differs.

The CLB achieves an overall OOS R^2 of only 0.023 (if we use common evaluation sample ($N = 2,973$), it even worse, only -0.07)—and is negative for the 10Y segment (-0.084) and at long horizons (18M: -0.058; 24M: -0.073)—far below ARX (0.374) and LightGBM (0.490). The gap between CLB and ARX is 0.351 overall, ranging from 0.330 (10Y) to 0.413 (3M tenor). These results confirm that the predictive gains of the full models do not simply reflect the linear relationship between the consensus forecast level and forecast errors. Rather, the gains arise from the broader feature set—including forecast revisions, U.S. interest rate expectations, and global financial variables—and the non-linear interactions among them.

The near-zero and negative CLB performance is itself informative: the consensus forecast level

alone has essentially no linear predictive content for subsequent forecast errors, and in the 10Y segment actively misleads a simple linear model. This implies that `bbg_consensus_tph`'s high SHAP importance in LightGBM (mean $|\text{SHAP}| = 0.44$, Section 5.3) reflects its role as a *non-linear* signal—the relationship between the consensus level and forecast errors is non-linear and context-dependent, requiring the interaction effects of gradient boosting to exploit—rather than a simple linear predictor that any model could capture.

6 Discussion

6.1 Why Does Machine Learning Work? Economic Interpretation

The central empirical finding of this paper is that all three machine learning models statistically significantly outperform professional consensus forecasts, with OOS R^2 ranging from 0.374 to 0.490. This section provides an economic interpretation of these results, addressing the question of *why* machine learning is useful in this setting. The CG (2015) diagnostic tests in Section 3.5 already confirm that consensus forecast errors contain statistically significant predictable components at short and long horizons, rejecting rational expectations. The machine learning results further reveal that this predictability stems from at least three interrelated sources:

1. **Systematic overreaction in consensus forecasts.** SHAP analysis identifies `bbg_consensus_tph` (the consensus forecast level) as the single most important feature for LightGBM (mean $|\text{SHAP}| = 0.44$), with a clear directional pattern: high consensus forecast levels tend to be associated with negative forecast errors (overprediction), while low consensus levels tend to be associated with positive errors (underprediction). This pattern is already foreshadowed by the CG (2015) diagnostic results in Section 3.5: the estimated $\hat{\beta} < 0$ indicates that forecasters tend to overshoot when revising upward—the revision magnitude exceeds what is warranted, causing future errors to reverse direction. This is fundamentally different from the underreaction (information rigidity, $\hat{\beta} > 0$) mechanism, where forecasters move in the right direction but not far enough. Machine learning can systematically identify and exploit this overreaction pattern—because it is not subject to the human cognitive biases that produce it—learning purely from historical data the statistical relationship between consensus levels and the direction of future errors, and generating corrections when the consensus overshoots.
2. **Non-linear transmission of U.S. interest rate expectations.** `us_2y_rate` (U.S. 2-year yield consensus forecast) ranks second in overall SHAP importance (mean $|\text{SHAP}| = 0.38$), with a consistent directional effect: high U.S. short-end rate expectations are associated with systematic underprediction of emerging-market sovereign yields, consistent with Jaramillo and Weber (2013). However, ARX substantially underperforms LightGBM at medium to long horizons, indicating that this transmission exhibits non-linear threshold structure—when U.S. rate expectations exceed certain levels, the response of emerging-market yields amplifies non-linearly. This threshold effect is precisely the structure that gradient-boosted decision trees are designed to capture, but that linear models cannot represent.
3. **Complex interaction effects of global risk sentiment.** `SPX_Index` (S&P 500) ranks third (mean $|\text{SHAP}| = 0.35$), with heterogeneous importance across maturities: its relative

importance is higher in the 10Y segment than in the 2Y segment, indicating that long-tenor sovereign yields are more sensitive to global equity market sentiment than short-tenor yields. The direction and magnitude of equity market sentiment’s impact on yield forecast errors varies across maturity-horizon combinations in ways that linear models cannot represent, while LightGBM captures these patterns through feature interaction effects.

Why is LSTM most useful in specific contexts? LSTM’s advantage is concentrated in the 2Y maturity, long horizons, and high-stress regimes. When global monetary policy enters a sustained tightening cycle, professional forecasters’ systematic underestimation of 2Y yields tends to persist across multiple consecutive quarters, generating positively autocorrelated forecast errors over time. LSTM’s memory mechanism can identify this pattern of “persistent mistakes,” while static LightGBM—though capable of capturing cross-sectional non-linearities—cannot fully represent these cross-period temporal dependencies. This is consistent with the “prestakes” concept of Bianchi et al. (2025): at market turning points, forecasters tend to anchor on recent trends and underweight mean reversion, and LSTM can learn this cross-period pattern from historical data.

Why does the linear ARX already perform well? ARX achieves an overall OOS R^2 of 0.374, indicating that consensus forecast biases have a “linear core”—primarily composed of the linear effects of the forecast revision and the consensus level itself, corresponding to the linear component of the overreaction pattern described by the CG (2015) framework. Machine learning builds on this foundation by capturing the non-linear interaction effects and temporal dynamics that ARX misses, generating the progressive improvement structure observed in this paper: ARX $\rightarrow$ LSTM $\rightarrow$ LightGBM.

6.2 When Does Machine Learning Work Best? A Maturity-Specific View

The results of this paper show that no single model dominates across all maturities, forecast horizons, and market conditions. This finding has a deeper economic interpretation: the three maturity segments correspond to three distinct market pricing mechanisms, and these three mechanisms happen to align with three different types of learnable structure.

The 3M sovereign yield is close to money market rates, primarily reflecting current monetary market conditions. The dominant source of forecast error is a linear overreaction pattern—forecasters overshoot when revising short-term rate expectations, but this overshoot is linear and predictable, already captured by the forecast revision term in ARX. The 2Y sovereign yield primarily reflects market expectations of the central bank’s future policy path. Monetary policy exhibits strong cross-period inertia: once a hiking or cutting cycle begins, it tends to persist across multiple quarters. Forecasters’ biases regarding the policy path therefore exhibit cross-period persistence—precisely the structure that LSTM’s memory mechanism is designed to capture. The 10Y sovereign yield, in addition to policy expectations, incorporates a term premium driven by non-linear shocks from global risk sentiment—the interaction between VIX, equity market conditions, and the consensus forecast level determines whether 10Y yields are over- or under-predicted, a static non-linear interaction structure that LightGBM is designed to capture.

This “three mechanisms, three models” framework elevates the paper’s empirical findings from “which model performs best where” to “why different maturities require different models,” providing

a unified theoretical interpretation for the stratified results that follow.

By maturity:

1. **10Y maturity: static non-linear structure dominates; LightGBM is strongest.** LightGBM's OOS R^2 (0.480) is nearly double that of ARX (0.245), and the DM test confirms the difference is statistically significant ($p < 0.01$). SHAP analysis reveals that the primary predictable source of 10Y forecast errors is the non-linear interaction between global risk sentiment (SPX_Index) and the consensus forecast level. LSTM's OOS R^2 in the 10Y segment (0.349) falls below LightGBM's, indicating that cross-period temporal persistence in 10Y forecast errors is relatively limited, and that the temporal memory mechanism adds limited marginal value in this setting.
2. **2Y maturity: temporal dynamics dominate; LSTM is strongest.** LSTM's OOS R^2 (0.794) substantially exceeds both LightGBM (0.496) and ARX (0.568), and the DM test confirms LSTM's improvement over ARX is statistically significant ($p < 0.01$). The 2Y segment exhibits the strongest cross-period temporal persistence in forecast errors in this paper's sample, and LSTM's memory mechanism directly captures this statistical pattern; the underlying economic mechanism warrants further investigation in future research. The most striking episode in the sample is the 2022–23 aggressive hiking cycle, in which the Fed raised rates by more than 500 basis points over 16 months—the fastest tightening in 40 years (Figure 7). During this cycle, consensus forecasters systematically underestimated the pace and persistence of rate increases across consecutive quarters, generating the kind of cross-period persistent forecast errors that LSTM's memory mechanism is designed to capture.

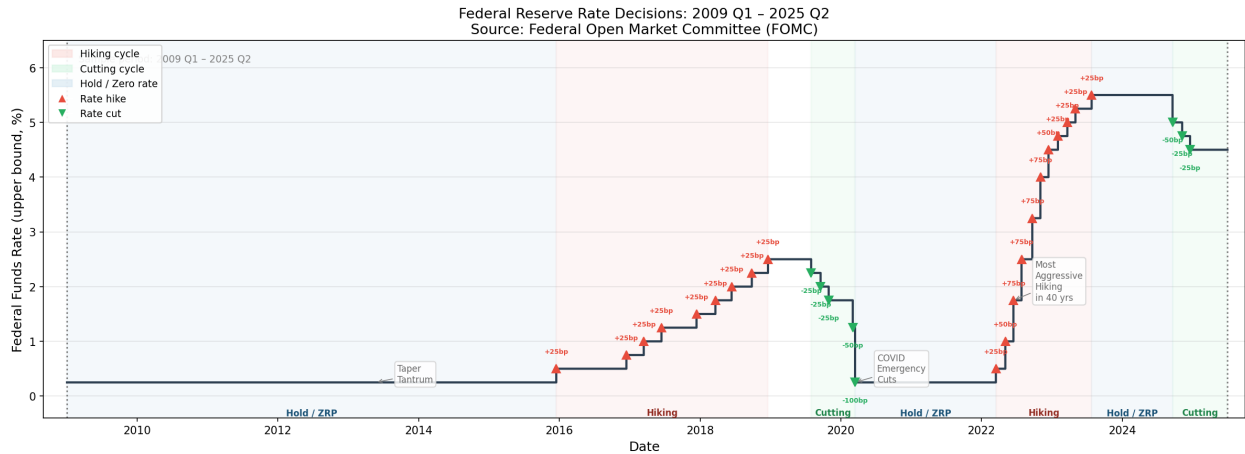


Figure 7: Federal Reserve Rate Decisions: 2009 Q1–2025 Q2. Red triangles (▲) indicate rate hikes; green triangles (▼) indicate rate cuts. Background shading denotes the prevailing monetary policy cycle: red = hiking, green = cutting, blue = hold/zero rate policy. Numbers indicate the size of each move in basis points. Source: Federal Open Market Committee (FOMC).

3. **3M maturity: linear patterns at short horizons, temporal dynamics at long horizons.** At the 3-month forecast horizon, ARX performs best, with the linear forecast revision term sufficient to capture the dominant predictable component. At long horizons (18M,

24M) and in high-stress regimes, however, LSTM substantially outperforms both ARX and LightGBM (3M tenor $\times$ 18M horizon: LSTM = 0.803 vs. LightGBM = 0.703), indicating that short-maturity yield forecast errors contain rich temporal dynamics at longer forecast horizons.

By forecast horizon:

4. **Short horizon (3M): linear patterns dominate; ARX is strongest.** Short-horizon forecast errors are predominantly governed by the linear overreaction pattern, which the forecast revision term in ARX already captures efficiently.
5. **Medium horizons (6M–12M): non-linear structure emerges; LightGBM is strongest.** LightGBM significantly outperforms ARX across this horizon range (DM test $p < 0.01$), indicating that medium-horizon forecast errors contain non-linear interaction effects beyond what the linear ARX can represent.
6. **Long horizon (18M): temporal dynamics are most critical; LSTM is strongest.** LSTM reaches its peak performance (OOS $R^2 = 0.706$), surpassing LightGBM (0.620), confirming that long-horizon forecast errors exhibit cross-quarter persistence that the LSTM memory mechanism is uniquely positioned to exploit.

By market condition:

7. **Moderate stress ($17.14 < \text{VIX} \leq 23.35$): optimal window for non-linear models.** LightGBM (0.548) and LSTM (0.532) achieve their highest OOS R^2 values in this regime, indicating that non-linear interaction effects among predictors are most active when global financial conditions are moderately elevated.
8. **Low stress ($\text{VIX} \leq 17.14$): ARX is relatively effective.** In tranquil global financial conditions, linear patterns dominate and ARX’s OOS R^2 (0.432) is comparable to LightGBM’s (0.435), with limited marginal gains from non-linear models.
9. **Extreme stress ($\text{VIX} > 23.35$): all models deteriorate; use with caution.** All three models decline substantially in extreme stress conditions, with ARX nearly vanishing (0.048), indicating that consensus forecasts themselves become highly unstable in extreme episodes and the regularity of forecast errors drops sharply.

Summary: a VIX-triggered model selection rule. Integrating the above findings, this paper proposes a practically implementable VIX-triggered switching rule:

- **$\text{VIX} \leq 17.14$:** the linear ARX is sufficient; trusting the consensus directly is also a reasonable choice.
- **$17.14 < \text{VIX} \leq 23.35$:** activate LightGBM or LSTM correction; this is the window of largest machine learning predictive gains.

- **VIX > 23.35:** all models face deteriorating performance; over-reliance on model correction may introduce additional uncertainty, and conservative use is advised.

In addition, on the maturity dimension: for 10Y long-tenor yields, LightGBM is the preferred model; for 2Y medium-tenor yields, LSTM delivers the strongest predictive gains; for 3M short-tenor yields, ARX is sufficient at short horizons while LSTM is preferable at long horizons. The real-time observability of VIX makes this switching rule operationally feasible, providing practitioners with a simple, empirically grounded model selection criterion.

6.3 Limitations and Future Research

Sample limitations:

- The panel covers 16 emerging markets over 2009–2025, with geographic coverage concentrated in Central and Eastern Europe and Southeast Asia. Latin America (only Brazil and Mexico) and Africa (absent from the sample) are underrepresented, which may limit the generalizability of the conclusions. Bulgaria (44 observations) and Turkey (43 observations) have very limited sample sizes and should be interpreted with caution.
- The sample begins in the post-GFC recovery period and does not include the core phase of the 2008 financial crisis, which may affect the representativeness of the extreme stress regime analysis.
- **Russian data reliability post-2022.** The sample includes Russia through 2025 Q2. However, following the imposition of broad-based international sanctions in February 2022, the reliability and representativeness of Russian sovereign yield data are subject to serious concerns, including bond market suspensions, withdrawal of major market participants, and restricted data accessibility. Results for Russia from 2022 onward should therefore be interpreted with caution. A robustness check excluding Russian observations from 2022 Q1 onward is a natural extension; given the limited weight of Russia in the equal-weighted primary specification (1 of 16 economies), we expect the impact on main conclusions to be modest.

Data limitations:

- The crisis dummy variables are never triggered during the sample period, requiring VIX quantiles as a substitute stress indicator. VIX proxies global financial stress rather than country-specific crisis events—the two are not economically equivalent. Future research could adopt more granular country-level stress indicators, such as sovereign CDS spreads or credit rating changes.
- `current_yield.t` has approximately 38% missing values, limiting its full utilization in the models.
- The predictor set consists entirely of numerical variables and does not incorporate high-frequency or real-time information. Forward-looking indicators such as the Citigroup Economic Surprise Index (ESI) and the five-year, five-year forward inflation breakeven rate may contain incremental predictive information not captured by the quarterly numerical feature set.

Model limitations:

- LightGBM uses time-series cross-validation for hyperparameter selection, while LSTM uses fixed hyperparameters—an asymmetric tuning strategy motivated by computational constraints. Future research could conduct systematic hyperparameter search for LSTM to assess the sensitivity of results to this design choice.
- This paper does not consider ensemble methods. Dynamically weighting the predictions of ARX, LightGBM, and LSTM according to prevailing VIX levels could potentially deliver further improvements in overall forecasting performance.
- The model set is deliberately kept parsimonious and does not explore a broader model space, including: Transformer architectures (whose self-attention mechanism may outperform LSTM in capturing long-range temporal dependencies); hybrid LightGBM-LSTM models (combining static non-linear structure with temporal memory); and regularized linear models such as Lasso and Ridge (as intermediate benchmarks between ARX and LightGBM).
- The predictor set does not incorporate textual information. Fed FOMC statements, meeting minutes, and Chair speeches contain substantial forward-looking semantic signals about future policy paths. Extracting hawkish/dovish orientation and policy uncertainty through tokenization and semantic analysis could provide incremental predictive information beyond numerical features. Incorporating NLP-derived features into the forecast error correction framework is a promising direction for future research.
- Major event shocks—such as geopolitical crises and surprise central bank decisions—are not explicitly modeled. VIX partially captures event risk, but cannot distinguish the heterogeneous impacts of different event types.

Directions for future research:

1. **Broader geographic coverage.** Extending the framework to a wider emerging market sample—particularly Africa and the Middle East—would assess the cross-regional generalizability of the conclusions.
2. **Dynamic model ensembles.** Exploring VIX-triggered ensemble strategies that dynamically adjust the weighting of ARX, LightGBM, and LSTM predictions according to real-time observable market conditions.
3. **Other asset classes.** Applying the forecast error correction framework to other asset classes—such as emerging market equities and exchange rates—to assess the broader applicability of machine learning in forecast correction settings.
4. **Economic mechanism of 2Y persistence.** Further investigating the economic mechanism underlying the strong cross-period temporal persistence of 2Y yield forecast errors, to provide a deeper explanation for why LSTM performs best in this maturity segment.

5. **Broader model exploration.** Systematically testing Transformer architectures, hybrid LightGBM-LSTM models, and regularized linear models to more comprehensively characterize the model space for emerging-market yield forecast correction.
6. **Textual information and NLP features.** Incorporating NLP-derived features from Fed FOMC statements and meeting minutes—such as hawkish/dovish orientation indices and policy uncertainty measures—into the predictor set, to explore the incremental predictive value of textual information for emerging-market sovereign yield forecast errors.

7 Conclusion

This paper examines whether machine learning can systematically outperform professional consensus forecasts out of sample, correcting systematic biases in emerging-market sovereign yield predictions. Using a panel dataset covering 16 emerging-market economies over 2009 Q1 to 2025 Q2, this paper compares three estimated models—ARX, LightGBM, and LSTM—against the professional consensus benchmark within a forecast error correction framework.

Main findings:

- **Consensus forecast errors contain statistically significant systematic patterns.** The CG (2015) diagnostic tests in Section 3.5 confirm that consensus forecasts exhibit significant overreaction at short (3M) and long (18M, 24M) horizons—forecast revisions negatively predict forecast errors ($\hat{\beta} < 0$), rejecting rational expectations. This provides the theoretical foundation for machine learning forecast correction.
- **All three machine learning models statistically significantly outperform the consensus benchmark.** ARX, LightGBM, and LSTM achieve overall OOS R^2 of 0.374, 0.490, and 0.470 respectively. Clark-West tests confirm that all three models outperform the consensus at least at the 10% significance level across almost all maturity and horizon subsamples, indicating that consensus forecast errors contain systematic, learnable patterns that can be identified and corrected using publicly available information.
- **Non-linear structure and temporal dynamics deliver progressive incremental predictive value.** LightGBM nearly doubles ARX’s OOS R^2 in the 10Y segment (0.245 to 0.480), with the DM test confirming the gap is statistically significant ($p < 0.01$). SHAP analysis attributes the gains primarily to non-linear interaction effects among the consensus forecast level, U.S. rate expectations, and global equity market sentiment. LSTM performs best in the 2Y segment (0.794) and at the 18-month horizon (0.706), reflecting cross-quarter temporal persistence in 2Y yield and long-horizon forecast errors—consistent with the historical record of consensus forecasters systematically underestimating the pace of rate increases across multiple consecutive quarters during the 2022–23 aggressive hiking cycle.
- **No single model dominates across all dimensions; the optimal model varies by maturity.** This finding has a deeper economic interpretation: the three maturity segments correspond to three distinct market pricing mechanisms. The 3M yield is close to money market rates, where linear overreaction dominates and ARX performs best. The 2Y yield

reflects policy path expectations, where cross-period temporal dynamics dominate and LSTM performs best. The 10Y yield incorporates a term premium subject to non-linear global risk sentiment shocks, where static non-linear interactions dominate and LightGBM performs best. The crisis analysis further reveals that machine learning’s predictive gains are largest in moderate stress regimes (VIX between 17 and 23), while all models deteriorate in extreme crisis conditions.

Theoretical and practical contributions:

- **Theoretical contribution.** This paper extends the belief distortions framework of Bianchi et al. (2022) to emerging-market sovereign yields, providing a micro-foundation for why machine learning outperforms consensus forecasts: the overreaction bias of professional forecasters constitutes a learnable statistical pattern, and machine learning can systematically identify and correct this pattern without being subject to the cognitive biases that produce it. The “three mechanisms, three models” framework further provides a unified theoretical interpretation for why different maturities require different model architectures.
- **Practical contribution.** This paper proposes a VIX-triggered model selection rule—activating non-linear ML correction when global financial stress is moderately elevated ($VIX > 17.14$) and using models conservatively under extreme stress ($VIX > 23.35$)—alongside a maturity-specific model selection map (LightGBM for 10Y, LSTM for 2Y, ARX for 3M at short horizons). The real-time observability of VIX makes this framework operationally feasible, providing practitioners with a simple, empirically grounded criterion for deploying machine learning in sovereign yield forecasting.
- **Methodological contribution.** This paper documents an explicit strategy for handling the severely unbalanced panel, provides a systematic comparison of rolling window configurations, and conducts Clark-West and Diebold-Mariano tests to confirm statistical significance—addressing what would otherwise be a fatal omission in a forecasting paper.

Outlook:

- **Broader model and data exploration.** Future research can extend the framework to Transformer architectures and hybrid LightGBM-LSTM models, incorporate NLP-derived features from Fed FOMC statements and meeting minutes, and integrate high-frequency forward-looking indicators such as the Citigroup Economic Surprise Index and inflation breakeven rates.
- **Broader geographic and asset class coverage.** Extending the framework to a wider emerging market sample—particularly Africa and the Middle East—and applying the forecast error correction approach to other asset classes such as emerging market equities and exchange rates would assess the broader applicability of the findings.
- **Fundamental role of machine learning in forecasting.** More fundamentally, the results of this paper suggest that in information-rich forecasting environments, the core value of

machine learning lies not in replacing professional forecasters, but in systematically identifying and correcting the inevitable cognitive biases in human forecasting behavior. This role is most valuable precisely when global financial conditions are most unstable and forecast errors are most systematic—the periods when getting the forecast right matters most.

A Data Construction Protocol

This appendix documents the complete data construction procedure for the forecasting panel, providing sufficient detail for replication by interested researchers.

A.1 Consensus Forecast Data: Identification and Extraction

The consensus forecast data used in this paper are drawn from the Economist Consensus Forecast (ECFC) module of a globally recognized financial database. This module provides a quarterly static snapshot for each forecast quarter, recording the consensus forecast value submitted by institutional contributors prior to the survey cutoff date. Each snapshot is indexed by a country-specific forecast code and target quarter (e.g., Brazil 2018 Q1).

The extraction procedure proceeds as follows. First, the consensus forecast code for each country-tenor-quarter combination is identified. Second, the corresponding forecast variables are extracted in batch using the Spreadsheet Add-in of the financial database. Third, Excel VBA scripts are used for initial data cleaning, including: removing redundant columns (CUSIP codes, source columns), extracting the tenor-specific ticker for each country-quarter, and standardizing column formats across countries.

A.2 Benchmark Bond Identification and Sample Consistency

The principal data construction challenge arises from the fact that the generic government yield index maintained by the financial database does not track a fixed bond instrument. Instead, it periodically rolls to a different actual bond as market liquidity conditions evolve—for example, consensus forecasts for Bulgaria’s 10-year sovereign yield may reference different actual bond instruments across quarters, even though the forecasting target remains consistently the 10-year sovereign bond of that country.

To ensure cross-period consistency, the following steps are taken for each country-tenor pair:

1. Using the relative timeframe view in the financial database, the actual bond instrument referenced by the generic yield index is identified for each quarter. Roll dates—quarters in which the benchmark bond changes—are recorded.
2. Based on the confirmed ticker for each quarter, the realized yield-to-maturity (YTM) of the corresponding bond at the target date is extracted.
3. The realized yield is matched to the corresponding consensus forecast to compute the forecast error $e_{i,m,t,h} = y_{i,m,t+h} - F_{i,m,t,h}^{\text{Con}}$.

A.3 Consensus Forecast vs. Realized Yield: Reconciliation

During data construction, a systematic discrepancy was identified between the consensus forecast value and the spot yield observed on the same date. This discrepancy arises because the consensus forecast snapshot records the average of institutional forecasts submitted prior to the survey cutoff, rather than the live market quote on any single date. Different institutions submit forecasts at different points within the quarter; the consensus value therefore reflects the information set

available during the submission window rather than at a precise single date. This paper adopts the convention of treating the forecast date as the quarter-end date of the survey cutoff, and uses the realized yield at the corresponding target date to compute forecast errors.

A.4 Merging of Additional Variables

Following the construction of the core forecast error panel, the following variables are merged using R:

- **Crisis indicators** (sovereign debt crisis, currency crisis, banking crisis): sourced from the financial database’s crisis database, merged on a country-year key.
- **Global financial variables** (VIX, Brent crude oil, S&P 500, gold, silver): extracted as weekly observations and aggregated to quarterly averages.
- **Exchange rates** (USD per local currency): extracted as weekly observations and aggregated to quarterly averages.
- **U.S. yield consensus forecasts** (3M, 2Y, 10Y) and U.S. macro forecasts: extracted from the ECFC module following the same procedure as Section A.1.

A.5 Sample Selection Criteria

The final usable sample is defined by the following criteria:

1. The target variable (forecast error) is non-missing: both the consensus forecast value and the realized yield are available.
2. The observation key (country $\times$ tenor $\times$ forecast date $\times$ horizon) is unique, with no duplicates.
3. The forecast date corresponds to a quarter-end date.

The raw candidate sample of 154,825 observations is reduced to 3,662 usable observations after applying `usable_consensus_flag` and `usable_target_flag` filters. Table 2 in the main text reports the distribution of usable observations across tenor and forecast horizon.

B Robustness — Alternative Weighting Schemes

This appendix reports the out-of-sample forecasting performance of all three models under two alternative weighting schemes: inverse frequency weighting (Robustness) and proportional weighting (Reference), alongside the equal weighting results (Primary) reported in the main text. The inverse frequency weighting scheme assigns higher weight to the underrepresented 2Y and 3M maturity segments, providing a robustness check on whether the main results are driven by the dominant 10Y segment. The proportional weighting scheme assigns weights proportional to each tenor’s share in the panel, serving as an additional reference.

The core conclusions are robust across all three weighting schemes: all models outperform the consensus benchmark, the progressive structure $\text{ARX} \rightarrow \text{LSTM} \rightarrow \text{LightGBM}$ is preserved, and LightGBM’s advantage in the 10Y segment and LSTM’s advantage in the 2Y segment are consistent. The overall OOS R^2 is slightly lower under inverse frequency weighting, reflecting the higher weight assigned to smaller, less stable subsamples.

Table 15: ARX: OOS R^2 under Alternative Weighting Schemes

	Equal Weight (Primary)	Inverse Freq. (Robustness)	Proportional (Reference)
<i>Panel A: Overall and By Tenor</i>			
Overall	0.374	0.277	0.339
10Y	0.245	0.006	0.274
2Y	0.568	0.642	0.427
3M	0.485	0.609	0.421
<i>Panel B: By Horizon</i>			
3M	0.557	0.527	0.469
6M	0.272	0.085	0.229
9M	0.342	0.239	0.314
12M	0.325	0.241	0.306
18M	0.363	0.242	0.372
24M	0.280	0.168	0.263

Note: OOS $R^2 = 1 - \text{MSE}_{\text{model}}/\text{MSE}_{\text{consensus}}$. $N = 3,437$ for all specifications.

Table 16: ARX (Inverse Frequency Weighting): OOS R^2 by Tenor and Horizon

Tenor	Forecast Horizon					
	3M	6M	9M	12M	18M	24M
10Y	0.181	−0.307	−0.081	−0.017	0.109	0.018
2Y	0.759	0.592	0.645	0.653	0.225	0.187
3M	0.406	0.476	0.673	0.611	0.652	0.609

Note: Negative values in red.

Table 17: LightGBM: OOS R^2 under Alternative Weighting Schemes

	Equal Weight (Primary)	Inverse Freq. (Robustness)	Proportional (Reference)
<i>Panel A: Overall and By Tenor</i>			
Overall	0.490	0.443	0.449
10Y	0.480	0.334	0.525
2Y	0.496	0.581	0.273
3M	0.519	0.594	0.517
<i>Panel B: By Horizon</i>			
3M	0.341	0.325	0.259
6M	0.449	0.352	0.356
9M	0.529	0.467	0.459
12M	0.525	0.441	0.476
18M	0.620	0.610	0.641
24M	0.529	0.485	0.543

Note: OOS $R^2 = 1 - \text{MSE}_{\text{model}}/\text{MSE}_{\text{consensus}}$. $N = 3,437$ for all specifications.

Table 18: LightGBM (Inverse Frequency Weighting): OOS R^2 by Tenor and Horizon

Tenor	Forecast Horizon					
	3M	6M	9M	12M	18M	24M
10Y	0.178	0.082	0.231	0.326	0.550	0.450
2Y	0.428	0.775	0.828	0.581	0.687	0.631
3M	0.174	0.296	0.636	0.685	0.752	0.533

Note: Negative values in red. None in this specification.

Table 19: LSTM ($L = 4$): OOS R^2 under Alternative Weighting Schemes

	Equal Weight (Primary)	Inverse Freq. (Robustness)	Proportional (Reference)
<i>Panel A: Overall and By Tenor</i>			
Overall	0.470	0.467	0.499
10Y	0.349	0.336	0.419
2Y	0.794	0.830	0.696
3M	0.733	0.744	0.690
<i>Panel B: By Horizon</i>			
3M	0.191	0.243	0.261
6M	0.321	0.365	0.400
9M	0.481	0.447	0.497
12M	0.496	0.487	0.533
18M	0.706	0.684	0.706
24M	0.528	0.503	0.534

Note: OOS $R^2 = 1 - \text{MSE}_{\text{model}}/\text{MSE}_{\text{consensus}}$. $N = 2,973$ for all specifications (observations with fewer than $L = 4$ historical periods excluded).

Table 20: LSTM $L = 4$ (Inverse Frequency Weighting): OOS R^2 by Tenor and Horizon

Tenor	Forecast Horizon					
	3M	6M	9M	12M	18M	24M
10Y	-0.257	0.354	0.417	0.438	0.623	0.379
2Y	0.962	0.287	0.459	0.560	0.810	0.796
3M	0.120	0.598	0.675	0.693	0.805	0.774

Note: Negative values in **red**.

C Bootstrap Confidence Intervals for OOS R^2

To assess the statistical uncertainty of the OOS R^2 point estimates reported in Section 5.5, we apply a block bootstrap procedure (Künsch, 1989) with block size $b = 8$ quarters (approximately two years) and $B = 1,000$ replications. The block structure accounts for temporal dependence in forecast errors. All results are reported on the common evaluation sample ($N = 2,973$).

Table 21: Block Bootstrap 95% Confidence Intervals for OOS R^2

	OOS R^2			
	Point Est.	Bootstrap Mean	SE	95% CI
<i>Panel A: Overall ($N = 2,973$)</i>				
ARX	0.374	0.264	0.058	[0.147, 0.371]
LightGBM	0.490	0.480	0.046	[0.384, 0.566]
LSTM	0.470	0.468	0.083	[0.286, 0.611]
<i>Panel B: By Maturity</i>				
ARX — 10Y	0.245	0.201	0.067	[0.059, 0.319]
ARX — 2Y	0.568	0.267	0.322	[-0.544 , 0.596]
ARX — 3M	0.485	0.401	0.088	[0.194, 0.513]
LightGBM — 10Y	0.480	0.457	0.046	[0.365, 0.547]
LightGBM — 2Y	0.496	0.593	0.159	[0.188, 0.749]
LightGBM — 3M	0.519	0.416	0.189	[-0.036 , 0.657]
LSTM — 10Y	0.349	0.341	0.104	[0.130, 0.528]
LSTM — 2Y	0.794	0.727	0.138	[0.407, 0.884]
LSTM — 3M	0.733	0.716	0.107	[0.451, 0.847]
<i>Panel C: By Horizon</i>				
ARX — 3M	0.557	0.345	0.252	[-0.421 , 0.581]
ARX — 6M	0.272	-0.114	0.141	[-0.424 , 0.121]
ARX — 9M	0.342	0.158	0.084	[-0.037 , 0.313]
ARX — 12M	0.325	0.242	0.068	[0.095, 0.361]
ARX — 18M	0.363	0.366	0.075	[0.202, 0.493]
ARX — 24M	0.280	0.253	0.073	[0.085, 0.371]
LightGBM — 3M	0.341	0.381	0.186	[-0.069 , 0.641]
LightGBM — 6M	0.449	0.223	0.119	[-0.025 , 0.431]
LightGBM — 9M	0.529	0.406	0.089	[0.214, 0.567]
LightGBM — 12M	0.525	0.516	0.082	[0.335, 0.653]
LightGBM — 18M	0.620	0.613	0.070	[0.460, 0.734]
LightGBM — 24M	0.529	0.512	0.054	[0.396, 0.610]
LSTM — 3M	0.191	0.070	0.552	[-1.266 , 0.753]
LSTM — 6M	0.321	0.322	0.174	[-0.040 , 0.608]
LSTM — 9M	0.481	0.488	0.113	[0.248, 0.683]
LSTM — 12M	0.496	0.495	0.107	[0.254, 0.670]
LSTM — 18M	0.706	0.700	0.082	[0.531, 0.826]
LSTM — 24M	0.528	0.523	0.131	[0.232, 0.737]

Note: Block bootstrap with block size $b = 8$ quarters and $B = 1,000$ replications (Künsch, 1989). 95% CI is the percentile interval. All results based on the common evaluation sample ($N = 2,973$). Point estimates for ARX in Panel A differ from those in Table 11 because the common sample excludes observations without LSTM predictions. Negative values in **red**.

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